Soil carbon gains and offsets under updated IPCC Tier 1 methods for organic soils, coastal wetlands and improved grasslands

Stock-change and flux-based GHG accounting of Irish country-level datasets

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Abstract

Since 2013, the IPCC Guidelines have allowed a country to report a soil carbon removal when it rewets drained peatland, restores a tidal marsh or mangrove, or improves grassland management. This study asked whether those removals are climate benefits once the methane and nitrous oxide emitted from the same land are counted. The default emission factors and reference stocks were encoded, and the emissions after rewetting were subtracted from those of continued drainage across thirteen cases. The gases were weighted at hundred-year and twenty-year global warming potentials, and the published 95 per cent ranges were propagated by Monte Carlo. Three problems were found. First, whether rewetting reduced or increased warming depended on the horizon chosen, not on the emission factors. Five of thirteen peatland cases changed sign between GWP100 and GWP20, and the same five fell below a 0.95 probability of benefit at GWP100. Second, the default reference stocks lay above the median of their source profiles in 34 of 37 cells, by 13 per cent at the median. A typical soil therefore held less carbon than its default. The printed range described the precision of a global mean, not the distance of any site from it. Third, Ireland's country-specific emission factors showed the defaults biased in opposite directions for carbon dioxide and methane on the same land. Therefore, this study suggests that soil carbon removals, as reported in national inventories, may not represent evidence of true climate benefits. Yet, IPCC guidance remains lacking on testing such outcomes separately.

1 Introduction

Since 2013, a country can report a soil carbon gain on three kinds of land at once under the IPCC methods, a rewetted peatland, a restored coastal wetland and an improved grassland, and no study has set those gains side by side on one greenhouse-gas-weighted axis. This paper assembles them, encodes the default tables that produce them, tests each gain against

the methane and nitrous oxide the same land emits at a hundred-year and a twenty-year horizon, and uses the factors Ireland has measured for the same pathways to test the defaults against a national dataset. The hypothesis is that the gains these methods allow a country to report are not climate benefits on a substantial share of pathways, for three reasons. National reporting weighs the gases at a single 100-year horizon and never sets a pathway's gain against its own methane and nitrous oxide. The gain and the penalty last for different lengths of time, and the printed uncertainty of the defaults cannot show how far they lie from a measured national value.

The methods are recent and unrevised, in that the 2013 Wetlands Supplement replaced Equations 2.24 and 2.26 of the 2006 Guidelines, which had "implicitly assumed that organic soils can only lose carbon", with an Equation 3.1 under which a rewetted organic soil removes carbon, the first negative Tier 1 emission factors in the history of the guidelines (Blain et al., 2014; IPCC, 2006). The same Supplement lets a restored tidal marsh or mangrove accumulate soil carbon "as long as the soil remains rewetted and vegetated", to a stock ceiling roughly two centuries away (Kennedy et al., 2014), and lets a rewetted wetland mineral soil recover its stock over forty years (Wickland et al., 2014). The 2019 Refinement kept the improved grassland factors of 1.14 and 1.17 that have stood since (McConkey et al., 2019; Ogle et al.; 2004), rebuilt the reference stocks they multiply from the profile compilation of Batjes (2011; Ogle et al., 2019), and answered its own peatland chapter with the words "No refinement", so every drained and rewetted factor in force today is the 2013 factor (Lovelock et al., 2019). Annex 1 to Volume 4 of the Refinement records those as the only changes to the two tables this paper runs, the reference stocks and the grassland factors (IPCC, 2019).

A gain reported under these methods is not the same thing as a benefit to the atmosphere, and the guidance gives no way to tell them apart. The Wetlands Supplement calculates each gas in its own equation and applies no greenhouse gas weighting anywhere in its 354 pages. The reporting rules of the Paris Agreement supply the weighting, requiring each country to add its gases at 100-year global warming potentials and to report the uncertainty of its totals

(UNFCCC, 2019). Other metrics are optional, and no rule asks for the net of a single pathway, its carbon dioxide removal set against the methane or nitrous oxide the same land emits. The choice of metric is itself a policy decision, which the IPCC methodology reports do not make and AR6 Box 7.3 declines to recommend (Forster et al., 2021). The gain and the penalty also last for different lengths of time, since coastal accumulation runs to a stock ceiling, inland mineral wetland recovery to forty years, mineral soil management to twenty and organic soil rewetting with no end, while the methane and nitrous oxide from the same hectares never stop. Petersson et al. (2024) define what a removal must satisfy before it counts as mitigation, a progressive increase in a site's stock over the pre-intervention condition by net transfer of carbon dioxide from the atmosphere, and none of it is satisfied by a default table alone. Wilson et al. (2016), written by the Supplement's own authors, did the subtraction that test requires at GWP-100 for inland organic soils and found large benefits for most land uses with one negative case, temperate nutrient-rich Forest Land, and Zhang et al. (2025) then showed by Monte Carlo on United Kingdom peatlands that the horizon alone turns one restoration step from a near certainty into a 42.2 per cent chance of increasing emissions under GWP-20, and their GWP* implementation is adopted in section 2.2. No study has taken the coastal, mineral wetland and grassland pathways through the same test, or asked how long each gain may be claimed against a penalty that never stops.

The defaults themselves are least certain in exactly the pools studied here. The rewetted temperate factors of Supplement Table 3.1 have 95 per cent bounds that span zero on carbon dioxide and start at zero on methane (Blain et al., 2014), the seagrass accumulation factor rests on six observations (Kennedy et al., 2014), and the boreal wetland mineral methane factor of Supplement Table 5.4 rests on one study with an interval as wide as its mean (Wickland et al., 2014). The 2019 Refinement warns that such defaults "can have inherently high uncertainties, particularly bias, when applied to specific countries" and that bias "is not necessarily captured in the uncertainty range" (Domke et al., 2019), because a printed range says how well a global average is known and not how far it lies from the soil being reported.

The only way to find that distance is to measure the soil, as Villarino et al. (2018) did in the Argentinean Semiarid Chaco by deriving country-specific stock change factors instead of propagating the defaults, and as Ireland has done for its organic soils by deriving Tier 2 factors from a national flux compilation (Aitova et al., 2023), which allows the default and the measurement to be compared on the same hectares.

Two findings result, the first being that on the organic soil pathways the sign of the benefit of rewetting is set by the emission metric and horizon rather than by the emission factor for five of thirteen land use, climate and nutrient combinations, and that the same five fail a 95 per cent probability threshold when the published bounds are resampled. The second is that on the mineral soil pathways the reference stocks every gain scales from lie above the median of their own source profiles in 34 of 37 cells. That excess reflects right-skewed profiles, and it does not show that national gains are inflated, since the mean is the right value for a national total. It matters for a parcel or project, where the typical soil holds less carbon than its default. The interval printed beside each stock is the precision of a global mean rather than the range a single parcel faces. The bias the 2019 Refinement warns about is the difference between a default and the land it is applied to. Ireland shows it directly, since its country-specific factors differ from the defaults in opposite directions for the two gases on the same land.

1.1 Terminology

Two distinctions apply throughout, the first being that flux-based accounting, area multiplied by an annual emission factor, applies to organic soils only. Mineral soil is instead stock-change, because Equation 2.25 of the 2019 Refinement differences two modelled equilibrium stocks and divides by a transition period (Ogle et al., 2019). There is no flux-based Tier 1 or Tier 2 method for improved grassland. Separately, there is no reference stock (SOC_{ref}) for peat, because Batjes excluded organic soils by design (Batjes, 2011). That exclusion is the structural reason the two accounting logics coexist and must be reported separately rather than summed.

2 Methods

The IPCC reports removals as negative and emissions as positive. For legibility, gains in this manuscript are reported as positive tonnes of carbon dioxide equivalent per hectare per year removed, and the sign is stated in every table caption. Sign conventions are treated as contractual, so that no quantity changes sign between a chunk and its caption.

2.1 Baselines

A rewetted organic soil emits methane and is a net source in absolute terms in every climate zone and nutrient class, which is a property of wet peat rather than a finding. The hypothesis concerns the *intervention*, so the quantity of interest is the difference between the rewetted state and the drained state it replaces, written as

$$\text{Benefit} = \text{Net flux}_{\text{drained}} - \text{Net flux}_{\text{rewetted}} \qquad (1)$$

Reporting the absolute rewetted flux as a result would invert the published conclusion of Wilson et al. (2016) and contradict Günther et al. (2020); the absolute fluxes therefore enter as model inputs only, and nowhere as findings.

2.2 Emission factors

Because the Supplement performs no weighting and AR6 recommends no single metric, results are reported under GWP-100 for comparability with inventory practice and under GWP-20 to expose the near-term methane penalty. Allen et al. (2022), signed by both sides of the GWP* dispute, recommend reporting short-lived and long-lived gases separately rather than choosing between metrics, and that position is followed here, so the set of metrics is a sensitivity rather than a claim about which metric is correct. The AR6 values used throughout are given in Table S1. The GWP* weighting of a methane flux E follows the implementation of Zhang et al. (2025) and is defined in *Equation 2*, under which a step change in a sustained flux is charged heavily for twenty years and a flux aged past that window is charged 0.28 of its GWP-100 value.

$$E_{CO_2we}(t) = [\,4.53\,E(t) - 4.25\,E(t-20)\,] \cdot GWP_{100} \qquad (2)$$

2.3 Reference mineral soils

Table 2.3 of the 2019 Refinement supplies the reference condition stock *SOCref*, 0 to 30 cm, under native vegetation (Ogle et al., 2019). The same table number in the Wetlands Supplement carries the drained methane factors, so the two are named by their document wherever both could be meant.

Of the 42 cells used here, 37 are the arithmetic means of the profile compilation of Batjes (2011), and the remaining five, for which that compilation had no data, take values retained from earlier guidance with a nominal error of ±90%. The cells span the Cool temperate, Warm temperate and Tropical zones, with stated 95 per cent errors from ±5% to ±90% of the mean; the Polar and Boreal rows are not included, and the consequence is set out in section 4.5. The stock change method those cells feed is Equation 2.25 of the 2019 Refinement, reproduced below in the form the guidance prints it (Ogle et al., 2019).

EQUATION 2.25 ANNUAL CHANGE IN ORGANIC CARBON STOCKS IN MINERAL SOILS

$$\Delta C_{Mineral} = \frac{SOC_0 - SOC_{(0-T)}}{D}$$

$$SOC = \sum_{c,s,i} \left(SOC_{REF_{c,s,i}} \cdot F_{LU_{c,s,i}} \cdot F_{MG_{c,s,i}} \cdot F_{I_{c,s,i}} \cdot A_{c,s,i} \right)$$

Where

$\Delta C_{Mineral}$ = annual change in organic C stocks in mineral soils, tonnes C yr^{-1}
SOC_0 = mineral soil organic C stock in the last year of an inventory time period, tonnes C
$SOC_{(0-T)}$ = mineral soil organic C stock at the beginning of the inventory time period, tonnes C
T = number of years over a single inventory time period, yr
D = time dependence of the stock change factors, the default period for transition between equilibrium SOC values, commonly 20 years, yr
SOC_{REF} = reference carbon stock, tonnes C ha^{-1}
F_{LU}, F_{MG}, F_I = stock change factors for land use, management and input, dimensionless
A = land area of the stratum, ha; c, s, i = climate zone, soil type and management system

2.4 Improved grasslands

The improved grassland pathway is Equation 2.25 run in the gain direction, and since *F_LU* is 1.0 for all grassland in all climates, the compound factor is *F_MG * F_I* and the whole gain reduces to one coefficient (McConkey et al., 2019; Ogle et al., 2019). The annual gain per hectare therefore reduces to *Equation 3*.

$$G = SOC_{\mathrm{REF}} \cdot \left(F_{\mathrm{MG}}^{\mathrm{improved}} \cdot F_{\mathrm{I}} - F_{\mathrm{MG}}^{\mathrm{start}}\right) \cdot \frac{44}{12} \cdot \frac{1}{D} \qquad (3)$$

Table 6.2 of the 2019 Refinement lists fertilisation as a qualifying improvement (McConkey et al., 2019), and fertilisation produces nitrous oxide, which is annual and open-ended, while the soil carbon gain is capped at D years by Equation 2.25. The penalty per kilogram of nitrogen applied combines the direct factor with the volatilisation and leaching pathways of Chapter 11 of the 2019 Refinement (IPCC, 2019) (*Equation 4*), and because the cumulative gain saturates at $D \cdot G$ while the penalty accrues without limit, the crossover year and the break-even rate follow from it (*Equation 5*).

$$p = \left[EF_1 + Frac_{\mathrm{GASF}} \cdot EF_4 + Frac_{\mathrm{LEACH}} \cdot EF_5\right] \cdot \frac{44}{28} \cdot \mathrm{GWP}_{100,\mathrm{N_2O}} \qquad (4)$$

$$t^* = \frac{D \cdot G}{N \cdot p/1000}, \qquad N_{\mathrm{break\text{-}even}} = \frac{1000 \cdot G}{p} \qquad (5)$$

2.5 Rewetted soil factors

Chapter 3 of the Wetlands Supplement applies from the year of rewetting with no transition period (Blain et al., 2014). Its factors are composite, covering soil and non-tree vegetation together, and are set out with their published bounds in Table S2. The balance the chapter reports is Equation 3.1, the equation that first allowed a rewetted organic soil to be reported as a removal (Blain et al., 2014).

EQUATION 3.1 NET GAINS OR LOSSES OF C RESULTING FROM THE BALANCE BETWEEN CO_2 AND CH_4 EMISSIONS AND REMOVALS

$$\Delta C_{\text{rewetted org soil}} = \mathrm{CO_2\text{-}C}_{\text{rewetted org soil}} + \mathrm{CH_4\text{-}C}_{\text{rewetted org soil}}$$

Where

$\Delta C_{\text{rewetted org soil}}$ = net C gain or loss in rewetted organic soils, tonnes C yr^{-1}

$\mathrm{CO_2\text{-}C}_{\text{rewetted org soil}}$ = net flux of CO_2-C, emissions or removals, from rewetted organic soil, tonnes C yr^{-1}

$\mathrm{CH_4\text{-}C}_{\text{rewetted org soil}}$ = net flux of CH_4-C, commonly emissions, from the rewetted organic soil, tonnes C yr^{-1}

Chapter 4 supplies the only factor in the guidance with no fixed end date other than a stock ceiling, applying “as long as the soil remains rewetted and vegetated”, through Equation 4.7 (Kennedy et al., 2014).

EQUATION 4.7 CO_2 EMISSIONS FROM REWETTING, REVEGETATION AND CREATION OF COASTAL WETLANDS

$$CO_{2\,SO\text{-}RE} = \sum_{v,s,c} (A_{RE} \cdot EF_{RE})_{v,s,c}$$

Where
$CO_{2\,SO\text{-}RE}$ = CO_2 emissions associated with rewetting, revegetation and creation activities by vegetation type v, soil type s and climate c, tonnes C yr^{-1}
A_{RE} = area of soil influenced by rewetting, revegetation and creation activities, ha
EF_{RE} = CO_2 emissions from aggregated mineral and organic soils influenced by rewetting and revegetation, tonnes C ha^{-1} yr^{-1}; EF_{RE} = 0 where no vegetation has been re-established or where re-establishment is expected by recolonisation

Chapter 5 supplies the only forty-year recovery in the guidance, as two consecutive twenty-year blocks, and the only stock-change gain in the Supplement, and it charges the methane of the rewetted state through Equation 5.1 (Wickland et al., 2014).

EQUATION 5.1 ANNUAL CH_4 EMISSIONS FROM REWETTED AND CREATED WETLANDS ON MANAGED LANDS WITH IWMS

$$CH_{4\text{-IWMS}} = \sum_{c} \left(A_{\text{IWMS},c} \cdot EF_{CH_4\text{-IWMS},c}\right)$$

Where
$CH_{4\text{-IWMS}}$ = annual CH_4 emissions from managed lands on inland wetland mineral soils where management has raised the water table to or above the land surface, kg CH_4 yr^{-1}
$A_{\text{IWMS},c}$ = total area of managed lands with mineral soil where the water table has been raised, in climate region c, ha
$EF_{CH_4\text{-IWMS},c}$ = emission factor for those lands in climate region c, kg CH_4 ha^{-1} yr^{-1}

Three scope caveats apply, the first being that the cultivated F_LU of 0.71 is defined for temperate and boreal regions only and is applied here to four tropical regions for want of a tropical value. Table 5.3 is a Cropland factor intended for use alongside management and input factors, set to unity here, and the boreal row, the narrowest margin of the nine, rests on a single study with a 95 per cent interval of ±76 on a mean of 76, so it should be read as indistinguishable from zero rather than as a demonstrated source.

2.6 Drained soil factors

Fourteen drained land use combinations are drawn from Chapter 2 of the Supplement (Drösler et al., 2014), of which thirteen are complete on all four gas terms; tropical Plantation acacia lacks a published nitrous oxide factor and is excluded from the benefit calculation. The Supplement provides no nutrient-status split for temperate Forest Land in any of these four tables, although Equation 2.7 has separate nutrient-rich and nutrient-poor terms; the equation

and the table contradict each other. The single temperate value is assigned to both nutrient classes here and recorded as an assumption rather than an IPCC default. This matters directly, because that row is the empirical anchor for the only published case in which rewetting does not deliver a benefit, and the rewetted factor it is differenced against has confidence intervals spanning zero on both carbon dioxide and methane. The methane term on the drained side applies Equation 2.6 of the Supplement in full, weighting the land and ditch factors by the ditch fraction (Drösler et al., 2014).

EQUATION 2.6 ANNUAL CH_4 EMISSIONS FROM DRAINED ORGANIC SOILS

$$CH_{4_organic} = \sum_{c,n,p} \left[A_{c,n,p} \cdot \left((1 - Frac_{ditch}) \cdot EF_{CH_4_land_{c,n}} + Frac_{ditch} \cdot EF_{CH_4_ditch_{c,p}} \right) \right]$$

Where

$CH_{4_organic}$ = annual CH_4 loss from drained organic soils, kg CH_4 yr^{-1}

$A_{c,n,p}$ = land area of drained organic soils in land-use category in climate zone c, nutrient status n and soil type p, ha

$EF_{CH_4_land_{c,n}}$ = emission factor for direct CH_4 emissions from drained organic soils, by climate zone and nutrient status, kg CH_4 ha^{-1} yr^{-1}

$EF_{CH_4_ditch_{c,p}}$ = emission factor for CH_4 emissions from drainage ditches, by climate zone and soil type, kg CH_4 ha^{-1} yr^{-1}

$Frac_{ditch}$ = fraction of the total area of drained organic soil occupied by ditches

2.7 Uncertainty propagation

Each factor was published as a central value with 95 per cent bounds that often straddled zero, so a point estimate could not settle the sign of a net flux. The benefit was therefore resampled by Monte Carlo, with triangular draws on the published value and bounds, and the consequences of that choice were examined in section 3.1. The draws covered the drained carbon dioxide, dissolved organic carbon, land methane, ditch methane and nitrous oxide factors, within the bounds printed in Supplement Tables 2.1 to 2.5 (Drösler et al., 2014). The carbon dioxide, dissolved organic carbon and methane factors of the rewetted state were sampled the same way from Supplement Tables 3.1 to 3.3 (Blain et al., 2014). Terms printed without a range, the ditch fraction among them, entered at their defaults.

2.8 Data sources

Every emission factor and reference stock in this analysis is transcribed directly from the published tables, and four external datasets supplement them.

Two are global, the first being the IPCC soil classification, a 5 arc-minute raster derived from the Harmonised World Soil Database, crossed with the FAO Global Ecological Zones of 2010 to give area weights by climate zone and soil class, with cell area computed on the sphere. The ecological zones are built from vegetation while the IPCC climate zones are climatic, so the match between the two zone systems is an approximation and the two mountain-system zones are ambiguous; a sensitivity reassigning them to the dry side is reported alongside the main result. Pathway areas come from the same database, 300,076 polygons, with Histosols mapped to the flux-based organic soil method and Gleysols to the wetland mineral method of Supplement Chapter 5 (Wickland et al., 2014); only dominant soil components survived the source merge, so the mapped areas are a floor rather than an estimate.

The Irish inputs are drawn together on one scale in *Figure 1*, because a stratification is easiest to audit against a country whose own soils are mapped independently. Both global layers are shown over the Republic in panels (a) and (b), where the ecological zone places the country in the cool temperate moist zone the reference stocks are keyed on while the soil classification resolves less cleanly, and the organic soil it finds is set beside the national soils map in panels (c) and (d). The carbon values attached to the national map are panel (e), and the coastal datasets behind the salinity threshold are combined in panel (f).

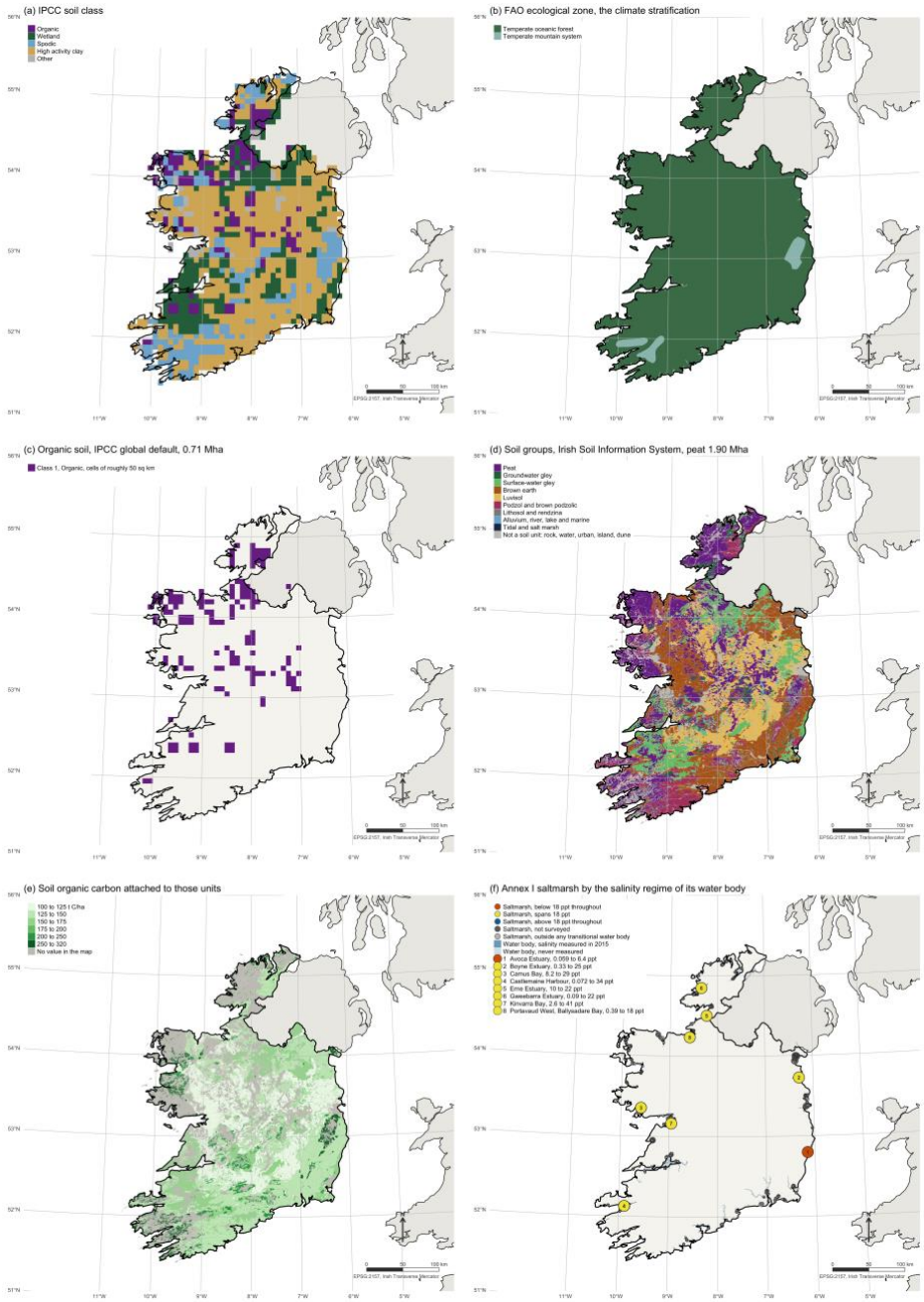


Figure 1: The Irish spatial inputs, six maps on one scale in EPSG:2157. (a) The 5 arc-minute IPCC soil classification derived from the Harmonised World Soil Database, clipped to the Republic of Ireland, which selects the soil class of the reference stock. (b) FAO Global Ecological Zones 2010, the climate stratification, which selects the climate zone. (c) Cells classified organic in the same IPCC layer, 0.71 Mha, the area to which the global default applies the flux-based method. (d) Soil groups of the Irish Soil Information System at 1:250,000, collapsed from its plain-English classes, with peat at 1.90 Mha and the two gleys marking the wetland mineral soils of the Chapter 5 method. (e) The soil organic carbon value attached to each unit of the same map, with unattributed area in grey. (f) Confirmed Annex I saltmarsh drawn at polygon centroids, sized by area and coloured by the salinity regime of the transitional water body containing it, over the Water Framework Directive transitional network shaded by whether Inland Fisheries Ireland measured it in 2015, with the eight measured systems numbered; Tralee Bay, the only system above 18 ppt throughout, is a coastal water body absent from this layer.

The other two are Irish, and the inventory analysis reads Ireland's 2026 submission to the UNFCCC, *IRL-CRT-2026-V1.0*, comprising 35 annual Common Reporting Table workbooks, from which implied emission factors are formed as reported emissions divided by reported area. The saltmarsh analysis joins the National Parks and Wildlife Service Saltmarsh Monitoring Project layer, 13,071 polygons, to the Environmental Protection Agency's Water Framework Directive transitional water bodies, 195 polygons, with salinity from Inland Fisheries Ireland's 2015 transitional-waters survey. That survey covers nine systems, of which eight appear in the transitional network; the ninth, Tralee Bay, is a coastal water body absent from the layer, which matters because it is the only system recorded above 18 ppt throughout. The salinity values are water-column measurements rather than marsh porewater, and are used as a proxy for it.

All processing code, together with the derived tables it produces, is archived with the manuscript and identified in the Data availability statement.

3 Results

3.1 Benefits of rewetting

Subtracting the rewetted flux from the drained state it replaces gives the quantity the guidance never computes. The drained side applies Supplement Equation 2.6 in full (Drösler et al., 2014), weighting land and ditch methane by *Frac_ditch*.

At the published point estimates every complete combination showed a benefit at GWP-100 (*Figure 2*; Table S3), reproducing the qualitative conclusion of Wilson et al. (2016) and Günther et al. (2020). The Monte Carlo below qualified that conclusion, because three of the thirteen medians fell below zero once the published bounds were sampled. At GWP-20 five combinations turned negative, meaning that rewetting those sites would increase radiative forcing over a twenty-year horizon. Every one was temperate or boreal, cases in which the drained carbon dioxide loss is modest while the rewetted methane flux is large. The tropical

rows, where drainage releases five to twenty tonnes of carbon per hectare per year, were insensitive to the metric because no plausible methane weighting can overtake a loss of that size.

Wilson et al. (2016) report one negative case, temperate nutrient-rich Forest Land at −0.25 t CO_2e ha^{-1} yr^{-1}, and that row, encoded here and complete on all four gas terms, did not reproduce as negative at GWP-100 but was one of the five that turned negative at GWP-20. The gap has two sources, the first being that Wilson et al. weight methane at 34 and nitrous oxide at 298, values that include climate-carbon feedbacks, against the AR6 values of 27.0 and 273 used here (Forster et al., 2021), and the second that the Supplement gives no nutrient split for temperate Forest Land, so the single published value is assigned to both classes, as recorded in section 2.6. The sign of this cell is therefore metric-dependent rather than settled, which illustrates the paper's central claim, because a disagreement between two studies about whether an intervention helps can be entirely an artefact of the weighting each chose.

Table 1: Monte Carlo on the rewetting benefit, 10,000 triangular draws from the published bounds, at GWP-100. Every term with a published range is sampled on both sides, and terms printed without one, the ditch fraction among them, are held at their defaults. `P(benefit > 0)` *is the probability that rewetting reduces net emissions.*

Land use	Zone	Nutrient	Median	5th percentile	95th percentile	P(benefit > 0)
Forest Land	Boreal	Poor	-0.44	-4.48	2.35	0.417
Forest Land	Boreal	Rich	0.21	-7.33	5.22	0.521
Forest Land	Temperate	Unsplit	-2.64	-15.89	6.60	0.348
Forest Land	Tropical	Unsplit	19.01	4.76	30.92	0.997
Cropland	Temperate	Rich	22.47	9.15	32.60	1.000
Cropland	Tropical	Unsplit	58.73	36.71	85.83	1.000
Grassland	Boreal	Unsplit	24.55	16.73	32.55	1.000
Grassland	Temperate	Poor	17.81	10.06	24.08	1.000
Grassland (deep-drained)	Temperate	Rich	14.22	0.77	23.93	0.960
Grassland (shallow-drained)	Temperate	Rich	1.86	-11.62	12.36	0.595
Grassland	Tropical	Unsplit	39.95	25.34	57.27	1.000
Peat extraction	Temperate	Unsplit	-2.70	-16.16	7.15	0.352
Plantation oil palm	Tropical	Unsplit	41.75	27.59	56.44	1.000

Two properties of the resampling bear on how *Table 1* should be read. The published value enters as the *mode* of the triangular, and the methane bounds are strongly right-skewed, so the sampled mean was well above the published default (Table S4), and it was that shift rather than the width of the sampling which moved three medians below zero when every point estimate was positive. Treating a 95 per cent interval as the full range of possible values also cuts off both tails, so the six rows printing exactly 1.000 should be read as “above 0.999”. Neither property disturbed the main result, since a normal and a uniform on the same bounds left the count failing the 95 per cent threshold at five and six of thirteen (Table S5), so that the identity of the rows was robust to the distributional assumption even though the individual probabilities are not.

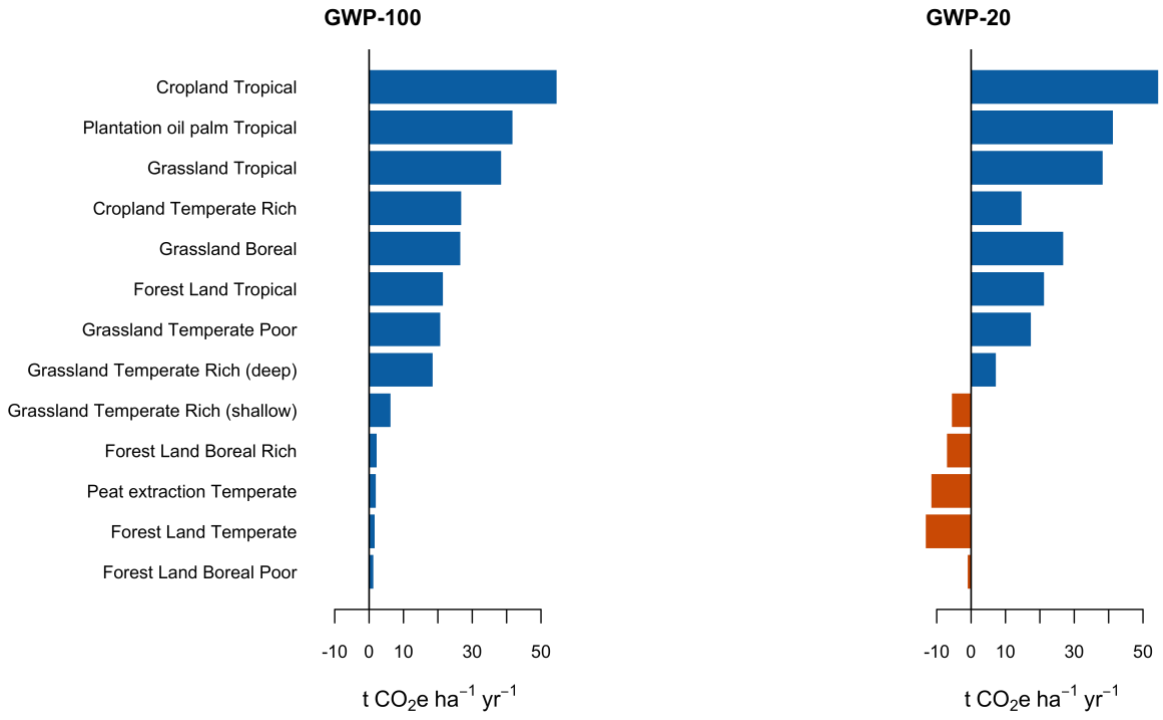


Figure 2: Benefit of rewetting by land use, climate and nutrient status, at two horizons. Bars left of zero are cases where rewetting increases radiative forcing. The GWP-20 panel is the same emission factors under a different weighting; nothing else changes between them.

The two tests selected the same rows, as *Figure 2* shows, since every combination turning negative at GWP-20 also failed the 95 per cent threshold at GWP-100, and no combination failed one test while passing the other. The agreement is expected, since the GWP-20 shift

and the Monte Carlo spread are both driven by the same rewetted methane term, so each statistic rises and falls with the size of the benefit relative to the methane the rewetted state emits. The agreement establishes not independent corroboration but the identity of the rows with no margin, five of thirteen pathways, Forest Land in the boreal zone at both nutrient levels and in the temperate zone, shallow-drained nutrient-rich temperate Grassland and temperate peat extraction, whose benefit is too small to survive any reasonable change in the inputs, and reporting only a GWP-100 point estimate, which is what inventory practice requires, conceals it either way.

3.2 Coastal wetlands

The salinity threshold was the decisive term for the *absolute* flux (Table S6), because below 18 ppt the single default of 193.7 kg CH_4 ha^{-1} yr^{-1} cost 5.23 t CO_2e ha^{-1} yr^{-1} at GWP-100, which exceeded the tidal marsh soil gain of 3.34, so on the Table 4.12 default a restored brackish tidal marsh was a net source in absolute terms, while the charge consumed most of the mangrove gain without extinguishing it. The sign is a property of that one default rather than a robust result, since the balance turned at an accumulation rate of about 1.43 t C ha^{-1} yr^{-1} and the independent syntheses discussed below put tidal marsh accumulation on the other side of that threshold. This is Tier 1 as published rather than a claim that brackish marshes are sources.

It does not follow that restoring the marsh is harmful, and the drained-baseline columns of Table S7 show why. Table 4.13 puts the drained state at 28.97 t CO_2 ha^{-1} yr^{-1}, so rewetting yielded a benefit of about 27 t CO_2e ha^{-1} yr^{-1} for tidal marsh and 30 for mangrove even under the freshwater methane penalty. The threshold changed the absolute flux by enough to reverse its sign while changing the benefit by less than a fifth, so that a restored wetland emitting methane remained a large improvement on the drained land it replaces.

What the threshold does determine is whether the pathway can be *credited* as a removal, since a scheme reporting the absolute soil gain and charging the methane would find the

balance negative, which matters for accounting and hardly at all for the atmosphere. Two eligibility conditions apply before it, namely that the site must be planted or seeded, unassisted recolonisation scoring zero, and that canopy cover must reach 10 per cent.

3.3 Inland wetlands

Inland wetland mineral soil rewetting was the clearest case in the guidance of a reported removal that is not a benefit (Table S8). The stock-change term was positive and would be reported as a removal under Equation 5.1, yet the Table 5.4 methane factor exceeded it in every climate region Table 5.2 publishes during the first twenty-year block, and in all but the boreal region during the second. Unlike the organic soil pathways there is no drained baseline in Chapter 5 to offset against, and the chapter states as much directly, that “Due to the lack of studies, however, we are unable to develop guidance for CH_4 emissions from drained IWMS at this time.” The comparison is therefore against the absolute flux and the conclusion correspondingly weaker, that the gain was smaller than its own offset rather than worse than the alternative.

3.4 Improved grasslands

Table S9 gives the resulting matrix, which ran from 0.92 to 6.71 t CO_2 ha^{-1} yr^{-1} from a nominal start and from 1.97 to 14.30 from a severely degraded one, accruing for twenty years and zero thereafter.

Table 2: Break-even nitrogen rate and crossover year. Beyond the crossover year the fertilised grassland is a net source in CO_2e terms.

Case	Gain (t CO_2 ha^{-1} yr^{-1})	Break-even rate (kg N ha^{-1} yr^{-1})	Crossover year (at 100 kg N ha^{-1} yr^{-1})	Crossover year (at 200 kg N ha^{-1} yr^{-1})
Lowest gain	0.92	156	31	From year 1
Median gain	2.98	506	101	51
Highest gain	6.71	1138	228	114

Because the carbon gain is finite and the nitrous oxide is perpetual (*Table 2*), at a penalty of 5.894 kg CO_2e per kg N applied, every fertilised improved grassland crosses to a net source

eventually, and the only question is whether the crossover falls inside the accounting horizon. The structure is the same one Lugato et al. (2018) identified for nitrogen-fixing cover crops, where roughly half of modelled sites became net greenhouse gas sources by 2060, and Li et al. (2005) found that nitrous oxide offsets 75 to 310 per cent of sequestered carbon in cropland over twenty years.

The penalty is avoidable, because fertilisation is only one of the qualifying improvements, and species improvement combined with grazing management receives the same F_{MG} factor with no nitrogen penalty at all, and biological nitrogen fixation is not a direct nitrous oxide source under the Guidelines, having been removed “because of the lack of evidence of significant emissions arising from the fixation process itself”. The grassland chapter that awards the gain contains no occurrence of nitrous oxide in its twenty-one pages, while the nitrous oxide methodology is in Chapter 11 of the same volume, so that a pathway can incur a penalty for which neither chapter is responsible.

3.5 Reference stocks bias

Every mineral soil gain above scales linearly with SOC_{ref}, so whatever is wrong with the reference stock passes undiminished into every removal built on it, and two different things can be wrong with it. The first is bias in the sense the 2019 Refinement uses the word, the distance between a global default and the true stock of the land it is applied to, which arises from how the profiles were chosen and cannot be seen from inside the sample. The second is the choice of which centre of a skewed sample to publish, which can be measured from the sample itself. Table 2.3 of the 2019 Refinement publishes arithmetic means (Ogle et al., 2019), while Batjes published medians with the median absolute deviation from the same profiles (Batjes, 2011, 2010), noting that “for skewed distributions the median is considered more robust than the mean since it is more resistant to erratic observations” (Batjes, 2011). The mean is nonetheless the right estimator of the total stock over a large area when the profiles represent that area, so the gap between the two centres is not in itself an error in a national

total. What it measures is how far the published value is from the typical profile, which is the quantity an analyst applies to a parcel and a project applies to a hectare.

The published mean exceeded the median in 34 of 37 cells, with a median excess of 13.0 per cent and a maximum of 154.5 per cent (Table S10), and in the three remaining cells, two volcanic and one wetland, the two centres coincided or the median was marginally higher. The largest single departures fell in the wetland and volcanic classes and in the tropical montane zone. Per-cell figures, however, treat a cell covering half a continent and one covering a few hundred thousand hectares as equally important, and weighting by the land each combination actually occupies asks a different question, namely whether an analyst applying Table 2.3 to a real landscape faces the gap often or rarely.

The weighting neither rescued the defaults nor condemned them (Table S11). Weighting by area pulled the mean excess down from 18.6 per cent to 15.5 per cent, so the larger combinations were slightly less skewed than the smaller ones, but the figure remained above the unweighted median of 13.0 per cent, and reassigning the ambiguous mountain zones moved it to 16.5 per cent. The gap is therefore not an artefact of a few small cells but is spread across the combinations that contain most of the world’s mineral soil, so that an analyst applying Table 2.3 to a real landscape faces it as a matter of course. Three caveats bound this result, namely that the zone matching is an approximation, that 20 per cent of classified land falls in the Boreal and Polar zones the table used here does not cover, and that about 265 Mha of organic soil is excluded by construction.

The interval printed beside each mean is a third quantity, and Table 2.3 defines it in its own footnote as the 95 per cent confidence limit, 1.96 standard errors expressed as a percentage of the mean. Recomputing that figure from the standard deviations and sample sizes of Batjes reproduced the printed column (Table S10), so the interval describes how precisely the mean of the sample is known, narrows as profiles are added, and says nothing about how far any one parcel lies from that mean. The spread of the profiles themselves, the standard deviation

as a share of the mean, had a median coefficient of variation of 54.9 per cent across the 37 cells against a median printed error of 13.4 per cent. The two differ by the square root of the sample size divided by 1.96, which is an identity rather than a finding, so the fault is not that the printed error is too small but that it is the wrong quantity for the use it is put to. An analyst who reads it as the uncertainty of the stock on a given parcel and propagates it through Equation 2.25 reports the precision of a global average, whereas a 95 per cent range for a single parcel at the median dispersion would run to about ±108%, eight times the printed figure.

The previous generation of defaults printed a different quantity again, and the change passed without comment. Batjes records that the 2006 Guidelines assumed a relative error of ±90% for every climate and soil cluster, “defined as 2xSD/MEA in said Guidelines”, which is a dispersion and corresponds to an assumed coefficient of variation of 45 per cent, and set against it the coefficients of variation observed in the profiles that replaced them, from 38 per cent to 106 per cent with a mean of 59 per cent (Batjes, 2011). The 2019 revision therefore did two things at once, replacing the profiles and replacing a dispersion with a precision, so the printed uncertainty on a typical cell fell from ±90% to about ±13%, roughly seven times narrower, while the spread of the soils behind it, 59 per cent by Batjes’s mean and 55 per cent by the median here, was wider than the 2006 table had assumed. Nothing in Table 2.3 or its footnotes marks the change of definition, and a reader setting the two tables side by side would take it for a gain in knowledge of the same size.

The bias the 2019 Refinement means is a separate matter, and Batjes states both of its sources rather than leaving them to inference. The 2006 defaults rested on roughly 3,500 profiles “with a strong geographical bias (77%) with respect to profiles from Brazil”, which the 2019 revision repaired by drawing on a wider selection, so the reference stocks have already been corrected once for a bias no confidence interval disclosed (Batjes, 2011). Both generations rest on profiles chosen by expert judgement rather than by a probability sample, of which Batjes writes that “there is a risk of bias as purposive sampling relies heavily on

personal (expert) judgement" and that "the only way to avoid bias inherent in purposive sampling is to follow probabilistic sampling" (Batjes, 2011). Neither source can be measured from the profiles, which is why no interval built from them can contain either, and a default with an unquantifiable sampling bias and printed with an interval describing only the precision of its own mean is the case the 2019 Refinement has in view when it warns that bias "is not necessarily captured in the uncertainty range" (Domke et al., 2019). The skew measured above is the part of the problem that can be seen, and it alone set the published value above the typical profile by 13 per cent at the median before any of the unseen part is counted.

One dimension of the same problem is not addressed here and should be. The whole Tier 1 stock method is defined over 0 to 30 cm, and Villarino et al. (2018) show for the Argentinean Semiarid Chaco that this "strongly underestimates CO_2 emissions" where disturbance reaches deeper, a limitation they overcome by modelling the vertical distribution of carbon to one metre from the surface stock. Batjes makes the same point about the reference condition, noting that the effect of a land-use change on the top 30 cm relative to the full metre "may vary markedly according to IPCC climate zone and soil class" (Batjes, 2010). The skew measured above therefore lies within a depth boundary that is itself too shallow for some of the interventions the defaults are applied to.

3.6 The assembled ledger

Table 3: The soil carbon gain ledger. Gains are maxima across climate and soil class; see the preceding sections for the full matrices.

Pathway	Source	Max gain ($t\ CO_2\ ha^{-1}\ yr^{-1}$)	Inventory period	Eligibility gate	Offset
Mangrove rewetting/revegetation, soil	WS Tb 4.12	5.94	To stock ceiling	Planted/seeded; >=10% canopy	CH_4 <18 ppt
Tidal marsh rewetting/revegetation, soil	WS Tb 4.12	3.34	To stock ceiling	Planted/seeded; >=10% canopy	CH_4 <18 ppt
Seagrass reestablishment, soil	WS Tb 4.12	1.58	No published ceiling	Planted/seeded	CH_4 <18 ppt
Rewetted organic soil, boreal nutrient-rich	WS Tb 3.1	2.02	From year 1, indefinite	Nutrient status	CH_4 + DOC
Rewetted organic soil, boreal nutrient-poor	WS Tb 3.1	1.25	From year 1, indefinite	Nutrient status	CH_4 + DOC
Inland wetland mineral soil, rewetting	WS Tb 5.3	4.95	40 years	None	CH_4
Improved grassland, nominal-improved + high input	2019 Tb 6.2	6.71	20 years	Improvement applied	N_2O if fertilised
Improved grassland, degraded-improved + high input	2019 Tb 6.2	14.30	20 years	Improvement applied	N_2O if fertilised

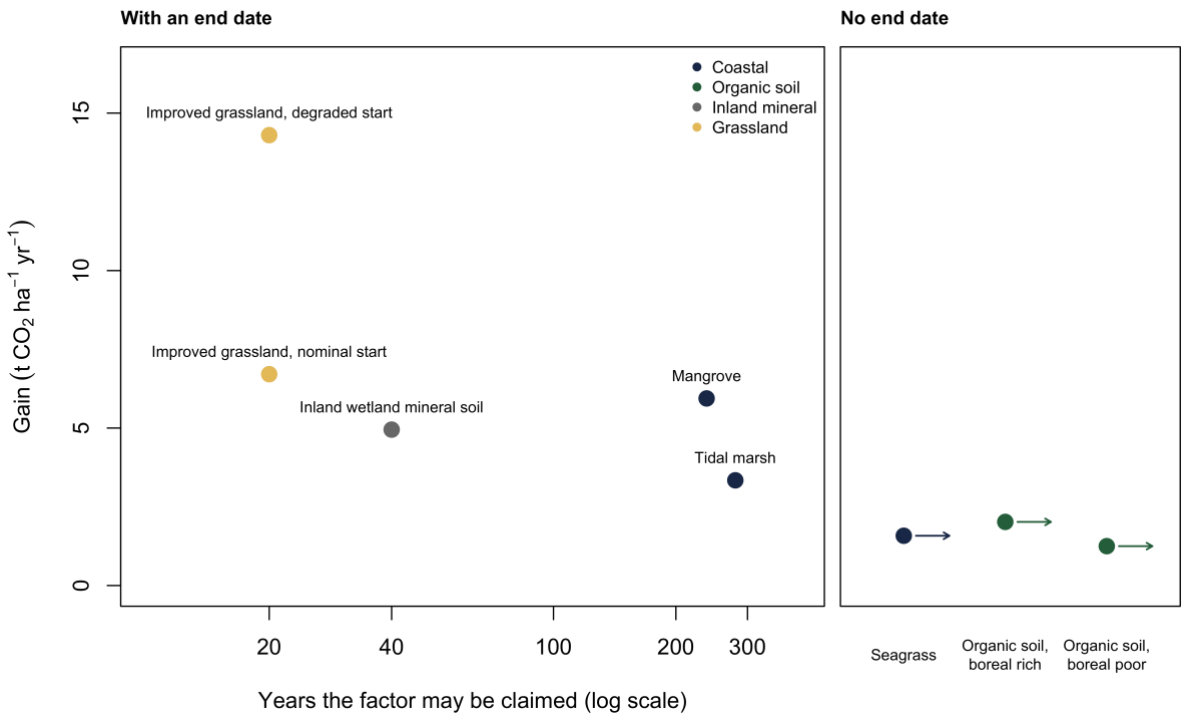


Figure 3: Annual gain against the years over which it may be claimed. The left panel holds pathways with an end date, on a logarithmic axis. Mangrove and tidal marsh are placed at the years a bare soil takes to reach its Wetlands Supplement Table 4.11 stock ceiling, an upper bound. The right panel holds the pathways with no end date in the guidance, rewetted organic soils and seagrass, which has no published ceiling. Two pathways with the same annual factor are not the same asset if one runs for twenty years and the other to a stock ceiling two centuries away.

The ledger, *Table 3* and *Figure 3*, makes the asymmetry visible, in that the pathway with the largest annual gain is not the one with the largest lifetime gain, because the inventory periods differ. A mangrove credited with 5.94 t CO_2 ha^{-1} yr^{-1} to a stock ceiling roughly two centuries away and an improved grassland credited with a comparable annual figure for exactly twenty years are not the same asset, yet nothing in the guidance requires the inventory period to be reported alongside the factor.

Everything above is expressed as a rate, stating what a hectare is credited with and for how long rather than whether the pathways are collectively worth a hundred megatonnes or ten gigatonnes. The land the rates apply to is mapped in *Figure 4*, and the scale calculations are set out in Tables S12 and S13, placed in the supplementary material because a total inherits every weakness of the rate and of the area at once. Three properties matter here, the first being that the organic soil total is an upper bound, not a projection, assuming every mapped

hectare is drained and then rewetted, and that it fell substantially between GWP-100 and GWP-20. No total is formed for the wetland mineral pathway, negative at scale as it was per hectare, because Supplement Chapter 5 applies to cultivated wetland mineral soil (Wickland et al., 2014) and no global layer resolves the cultivated fraction of Gleysols. The grassland figure is an upper bound on an upper bound, counting steppe nobody manages and charging no nitrous oxide, so the pathway with the largest apparent magnitude has the least defensible denominator.

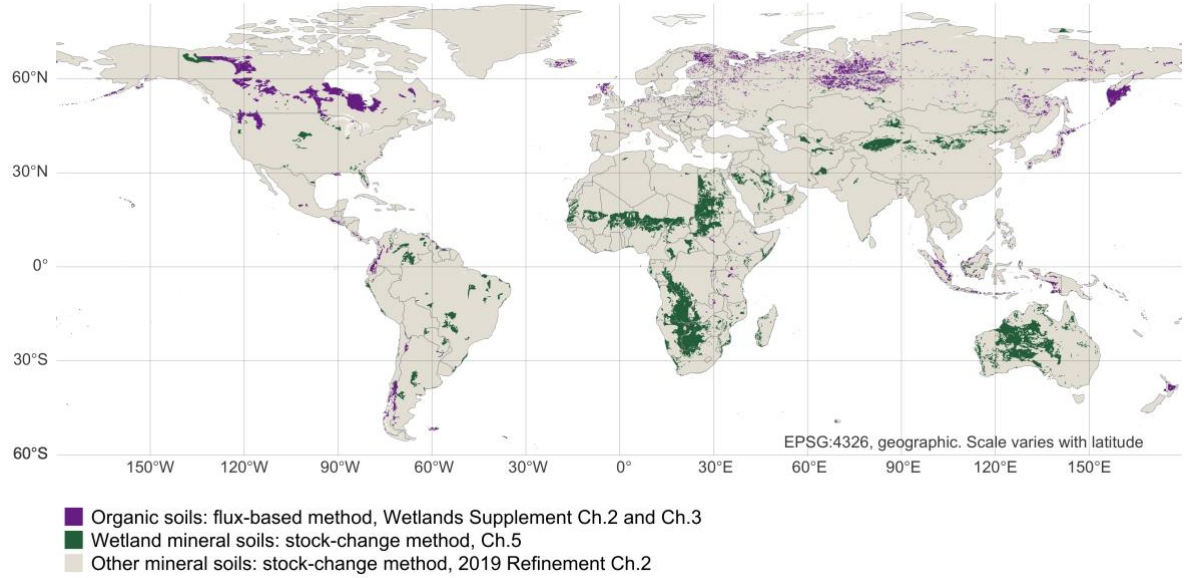


Figure 4: The soils the Tier 1 gain pathways apply to, from the 5 arc-minute IPCC soil classification. Organic soils carry the flux-based method of Wetlands Supplement Chapters 2 and 3; wetland mineral soils carry the Chapter 5 stock-change method; all other mineral soils fall under the 2019 Refinement Chapter 2 reference stocks. The two coloured classes are the soils this paper's ledger covers, and the boundary between them is the boundary between the two accounting logics of the guidance.

The mapped areas are floors rather than estimates, and the resolution compounds the merge, since over Ireland the 5 arc-minute layer found 0.71 Mha of organic soil where the national soils map records 1.90 Mha of peat, missing the western seaboard and most of the midland raised bogs because a cell of roughly fifty square kilometres cannot resolve a bog of two (panels (c) and (d) of *Figure 1*). The undercount fell precisely where the flux-based method has its largest factors, and the Irish ratio calibrates the floor without correcting it.

3.7 Irish national case

Ireland has done the work the guidance does not require, having derived country-specific Tier 2 factors for organic soils from a national flux compilation (Aitova et al., 2023) and adopted them into its inventory. Where a national value exists, the paper's question can therefore be asked twice, once with the IPCC default and once with the measured factor.

Nutrient-poor drained grassland was the clearest case in the comparison (Table S14), its carbon dioxide factor roughly four times *below* the default while its methane factor was more than six times *above* it, on the same hectares of the same land use. Aitova et al. (2023) attribute the first to a Tier 1 pool dominated by intensively managed continental grassland and the second to a pool of deep-drained sites, against Irish conditions that are shallow-drained and oceanic. A single default cannot be conservative for both gases at once, and widening the confidence interval does not repair it, because the two errors point in opposite directions.

The Supplement compounds this by offering no cell to fall back on, because Supplement Tables 2.1, 2.3 and 2.5 separate deep from shallow drainage (Drösler et al., 2014) only within *nutrient-rich* temperate grassland. There is no Tier 1 entry for shallow-drained nutrient-poor temperate grassland, which is the commonest Irish case.

The strongest test used nothing but what Ireland reported to the UNFCCC. Ireland's tables split the balance for a hectare of organic soil between Table 4.D, the on-site carbon stock change, and Table 4(II), the off-site carbon dioxide, nitrous oxide and methane. Both entered the national total at 100-year global warming potentials (UNFCCC, 2019), but neither table reported the benefit, the difference between the rewetted and the drained state.

Table 4: Industrial peat extraction in Ireland, 2024, t CO_2e ha^{-1} yr^{-1}, positive = net source. Built entirely from Ireland's reported implied emission factors: on-site soil carbon from Table 4.D plus off-site CO_2, N_2O and CH_4 from Table 4(II).

Metric	Drained	Rewetted	Benefit of rewetting
AR5 GWP-100	15.02	8.19	6.83
AR6 GWP-100	15.00	7.98	7.02
AR6 GWP-20	16.19	19.08	-2.89

The sign flipped with the horizon inside a live national inventory (*Table 4*), where rewetting Irish industrial peat extraction was worth about seven tonnes of carbon dioxide equivalent per hectare per year at GWP-100 and cost nearly three at GWP-20. The driver was a rewetted methane implied factor around nine times the drained one, which a hundred-year weighting discounts and a twenty-year weighting does not.

One boundary difference is worth naming as the likeliest source of disagreement with any factor-based Irish estimate, which is that the figures here are the full reported balance and the off-site terms of Table 4(II) are large enough on these pathways to move the sign on their own. Whether a national pathway helps or harms can therefore turn on where the inventory boundary is drawn, before any question of metric arises.

The near-natural row made the same point without any intervention at all. Ireland reports 885,000 hectares of near-natural peatland as a soil carbon removal, and it is one, yet adding the dissolved organic carbon and methane Ireland reports for the same hectares turned it into a net source of about 1.4 t CO_2e ha^{-1} yr^{-1} at AR6 GWP-100, roughly 1.2 Mt nationally, so the reference condition of restoration does not net out its own methane.

The coastal result turns on a single threshold, and since no dataset joins Irish marsh extent to salinity and both layers are public, the join was built here.

Two results followed, set out in Table S15 and panel (f) of *Figure 1*, and the second is the larger. About a tenth of Ireland's confirmed Annex I saltmarsh lay in estuaries recorded below 18 ppt in the 2015 survey, and on those hectares the Supplement Table 4.14 default (Kennedy

et al., 2014) cost more than the soil gain was worth. Not one hectare classified above the threshold throughout, and the reason is a matter of data rather than ecology, since the only system recorded above 18 ppt for its whole range was Tralee Bay, a coastal water body absent from the transitional network the marshes are joined to. The saline side of the threshold is unreachable through this pairing of datasets rather than unoccupied.

The second result is that ninety per cent could not be classified at all, half of it lying in transitional water bodies where salinity has never been measured and two fifths outside the transitional network entirely. A threshold that determines the sign of a national pathway is therefore unresolvable for nine tenths of the resource using the data the State collects.

The same panel shows the datasets behind that classification, and the coverage is the point, because the confirmed Annex I stratum is a minority of the mapped saltmarsh, sharing the coast with desk-mapped potential polygons and with invasive *Spartina* sward that a Tier 1 reading would credit identically, and nine of the 195 transitional water bodies have a salinity measurement, scattered around the coast rather than chosen to represent the marsh resource. A threshold that decides the sign of a national pathway is being resolved from a survey designed to count fish.

Three limitations bound this, the first being that the salinities are water-column measurements rather than marsh porewater, which is generally fresher, and the second that they are single-survey ranges rather than annual means, which is why every classified hectare fell in the “spans” category. No Irish saltmarsh has been planted or seeded, so none of this area would qualify for a factor in any case, and the figure therefore bounds the exposure a future Irish pathway would inherit rather than a current liability.

The reference-stock logic excludes organic soils by design, and Ireland’s national soils map reproduces that exclusion in the dataset an analyst would actually open. The figures below are a scoping estimate, establishing the size of the gap rather than what should fill it.

559 The map states no soil organic carbon value for 32 per cent of the mapped area, and for 92
560 per cent of the peat, and the gap was not scattered (panel (e) of *Figure 1*). Mineral Ireland is
561 attributed almost everywhere, in a narrow band from roughly 100 to 200 t C ha^{-1}, while the
562 peat is blank in a single continuous absence covering the western seaboard, the north-west
563 and the midlands, so this is not a map with missing values but a map built on a stock-change
564 logic, declining to state a stock for soils that logic was never constructed to describe, and the
565 blank is a property of the accounting rather than of the survey.

566 Two questions follow, and neither can be settled here, the first being whether the absent
567 attribute is recoverable, given that the gap falls almost exactly on the soils to which the flux-
568 based method applies. The second must be resolved before any Tier 2 stock is drawn from
569 this map, because the map reports carbon over the mapped profile depth while Table 2.3 of
570 the 2019 Refinement is defined for 0 to 30 cm, so the two are not comparable until the depth
571 basis is reconciled, and that discrepancy is flagged here rather than adjusted for.

572 Ireland has measured its own direct nitrous oxide factor by fertiliser formulation. Applying
573 those factors to the crossover calculation of section 3.4, on the 2019 Refinement Table 2.3
574 default stock for Ireland's climate zone and dominant soil class, keeps the figure traceable to
575 an encoded default.

576 On that stock, Ireland's dominant mapped class being high-activity clay, the default
577 improvement gave 2.08 t CO_2 ha^{-1} yr^{-1} for twenty years, and at 180 kg N ha^{-1} yr^{-1} the crossover
578 fell in year 29 under calcium ammonium nitrate and in year 70 under protected urea.

579 The result points at a research question rather than settling one. If formulation rather than rate
580 is the dominant term, the nitrogen penalty attached to an improved-grassland gain is a policy
581 variable rather than a property of the pathway, and the crossover year becomes something a
582 jurisdiction can move. Two things would have to be established first, because the emission
583 factor rises with application rate (Rahman et al., 2021), so that a crossover computed at a
584 constant factor is conservative by an unquantified margin, and because the soil gain here is

585 the Tier 1 default, whereas a country with a lower modelled estimate for the same
586 management may find the penalty exceeding the gain from the first year, with no crossover at
587 all.

588 The metrics reported so far have been two horizons of the same pulse-based metric. GWP*
589 is constructed for a different question, namely how a *step change* in a sustained short-lived
590 flux, which is exactly what rewetting is, translates into warming.

591 GWP* reversed the ranking (Table S16), in that the same Irish methane flux which was heavily
592 charged at GWP-20 was charged 28 per cent of its GWP-100 value beyond the first two
593 decades, because a constant emission rate adds no further warming once established, so the
594 transition is charged heavily for twenty years and then almost not at all.

595 ## 4 Discussion

596 Two failures run through these results, the first being that the guidance cannot tell a reported
597 gain from a benefit, since the organic soil results show five of thirteen combinations changing
598 sign between GWP-100 and GWP-20, and five of thirteen failing a 95 per cent probability
599 threshold at GWP-100 alone. They are the same five, being Forest Land in the boreal and
600 temperate zones, shallow-drained nutrient-rich temperate Grassland, and temperate peat
601 extraction. Metric sensitivity and sampling uncertainty are not independent tests here; they are
602 two ways of detecting the same thing, which is that these five pathways have no margin, and
603 an analyst reporting the GWP-100 point estimate, which is what UNFCCC inventory practice
604 requires, discloses neither.

605 This bears directly on a live and unresolved disagreement, in that Günther et al. (2020)
606 conclude that rewetting drained peatland delivers an immediate climate benefit. Ojanen and
607 Minkkinen (2020) put the switchover at forty to a hundred years for forestry-drained sites, Ma
608 et al. (2024) at 110 years for inland marsh, Launiainen et al. (2025) find boreal peatland forests
609 warming for sixty to eighty years, and Schuster et al. (2024) report 525 years in global meta-

analysis and conclude that restoration “does not meet the timeframe set by the Paris Agreement”. The present results suggest that part of the spread is definitional rather than empirical, since the same emission factors yield a benefit or a penalty depending on the horizon at which methane is weighted, and the studies do not share a horizon.

The Irish inventory result is the sharpest form of this, since no factor changes between the two lines of *Table 4* and only the horizon does. A jurisdiction can therefore make a defensible mitigation claim and a defensible warming claim about the same hectares in the same year, and the guidance offers no basis for preferring either.

This has a direct implication for national reporting, in that Ireland reports at AR5 GWP-100, the metric under which its peatland accounting appears most favourable. There is no suggestion that the choice was made to obtain that result, since it is the reporting convention and was adopted for comparability. The difficulty is that a convention adopted for comparability is doing substantive work, and the required reporting does not show it, since paragraph 37 makes every other metric optional (UNFCCC, 2019).

Two consequences follow, the first being that metric choice should be declared as a methodological parameter with a stated horizon and scope rather than left to a default, and the second that where a result changes sign across the panel, the full panel should be reported rather than the member of it that supports the intended conclusion.

The reversal under GWP* in section 3.7 is not a reason to prefer that metric, but the reason the panel is reported rather than a winner declared. The three metrics answer three different questions, being cumulative forcing over a century, over two decades, and *additional* warming relative to a continuing baseline. A pathway beneficial under the first, harmful under the second and close to neutral under the third has been shown only to be metric-dependent, which is a statement about the accounting rather than about the peatland.

One caveat belongs with any use of GWP* here, in that it rewards a step down relative to a recent baseline, so it credits a jurisdiction for the historical fact of having drained its peatlands

and then stopping, and yields nothing to a country with little drained peatland to rewet. That is defensible as physics and difficult as policy, and it is why Allen et al. (2022) recommend reporting the gases separately rather than choosing between metrics.

The second failure lies in the defaults themselves, and it is not a failure of precision. The reference stocks lie above the median of the profiles they were drawn from in 34 of 37 cells, by a median of 13.0 per cent. Where the profiles represent the land, the mean is the right value for a national total, so this skew does not show that national gains are inflated. It matters for a parcel or a project, where the typical soil holds less carbon than the default. The interval published beside each default describes how well a global average is known, not how far it lies from any particular hectare. It replaced a dispersion in the 2019 table without notice, and it is the only measure of uncertainty the guidance provides. An analyst following the method exactly therefore cannot see how far a parcel lies from its default. The 2019 Refinement warns that bias "is not necessarily captured in the uncertainty range" (Domke et al., 2019). The Irish factors show that kind of bias directly, because the defaults lie above the Irish value for carbon dioxide and below it for methane on the same hectares.

The direction of the Irish differences matters more than their size. A single interval can be widened until it covers a factor that is too high or one that is too low, but it cannot cover both at once, and the Irish nutrient-poor grassland cell has a carbon dioxide factor roughly four times below the default while its methane factor is more than six times above it. No propagation of the published errors would reveal that, because propagation assumes the central values are unbiased and asks only how far they might scatter.

4.1 Drained-rewetted binary constraints

The Tier 1 method assigns one factor to "drained" and one to "rewetted" with no transition between them, and three independent lines of evidence say this is not how the systems behave. Evans et al. (2021) show that every 10 cm reduction in effective water-table depth cuts GWP-100 warming by at least 3 t CO_2 ha^{-1} yr^{-1}, making the flux a continuous function of

a variable the tier structure splits into two classes. The step assumption also ignores a transient, since Kalhori et al. (2024) observe methane peaking at 537 kg ha^{-1} yr^{-1} in year three and falling to 120 by year fourteen, matching IPCC defaults only thirteen to sixteen years after rewetting, while Vanselow-Algan et al. (2015) find methane still two to five times the default thirty years on, and the two datasets disagree about whether the transient ends by more than the difference between the defaults being applied.

The third line is spatial rather than temporal, and it operates at the scale at which a factor is actually applied. Phillips and Beeri (2008) measured six vegetation zones, ordered by wetness, across replicate prairie pothole basins in North Dakota and found methane falling from 277,800 micrograms per square metre per day in deep marsh to −730 in pasture, a change of sign within a single wetland basin, while nitrous oxide varied more within zones than between them. A polygon assigned one emission factor therefore averages a gradient running from a strong source to a small sink, and nothing in the estimate discloses that it has done so. This is the same defect as the water-table one, expressed in space rather than depth.

The direction of the resulting bias is not neutral, because the transient is a methane overshoot, so a step-change factor understates near-term warming, which is precisely the term the GWP-20 results showed to be decisive.

A third state lies outside the drained or rewetted classification altogether, in that the guidance recognises a drained soil and a rewetted one, both treated as stable, while a soil that dries and rewets repeatedly behaves like neither. Gao et al. (2016) incubated alpine wetland soil for 120 days and found that cycling between 60 and 100 per cent of water-holding capacity accelerated carbon mineralisation and nitrogen immobilisation relative to soil held saturated, with litter addition amplifying both. Crossing three cycle frequencies with three watering intensities in wetland microcosms, Gao et al. (2018) found methane emission highest at the highest frequency under the two wetter treatments and nitrous oxide highest at the highest frequency under the driest, so frequency and intensity interact rather than add. An incubation

and a microcosm are not a peatland, and both results are reported here as a mechanism rather than a magnitude. The mechanism matters because a rewetted site whose water table fluctuates is charged the rewetted factor regardless of how often it dries.

The dissolved organic carbon term rests on a similar assumption, and it can be tested directly against the encoded factors. Supplement Table 3.2 gives a rewetted flux of 0.08, 0.24 and 0.51 t C ha^{-1} yr^{-1} in the boreal, temperate and tropical zones (Blain et al., 2014) against drained values of 0.12, 0.31 and 0.82 in Table 2.2 (Drösler et al., 2014), so rewetting is credited with a reduction in every zone. Schwalm and Zeitz (2015) found the opposite over two years in lysimeters containing three northeastern German fen histosols, reporting that “DOC release of long-lasting ‘rewetted’ lysimeters was not lower than those from drained ones” and that the extent of release was determined by the degree of decomposition and by pH, “whilst the SOC content and water table depth were of lesser importance”. Their sites are one peat type in one region, and they cite long-term observations in which concentrations do fall to near-natural levels within ten to twenty years of rewetting, so this is not evidence that the default is wrong everywhere. It is evidence that the reduction is tied to drainage status while the variables that predict it are decomposition and pH, neither of which enters the tier structure at all.

Removing the reduction changed less than the argument might suggest. Setting the rewetted dissolved organic carbon equal to the drained value in each zone lowered the benefit by between 0.15 and 1.13 t CO_2e ha^{-1} yr^{-1}, and no combination changed sign at either horizon. The term was nonetheless 12 per cent of the smallest surviving benefit at GWP-100, boreal nutrient-poor Forest Land at 1.26 t CO_2e ha^{-1} yr^{-1}, which is one of the five rows the Monte Carlo already places below the 95 per cent threshold. A term too small to overturn a result on its own can still consume an eighth of the margin on the rows that have least of it.

Independent evaluations have moved several of the factors used here. He and Roulet (2023) revise the peat extraction factor down 12 per cent to 2.46 ± 0.25. On the coastal side, the tidal marsh accumulation default of 0.91 t C ha^{-1} yr^{-1} is two to three times below independent global

714 syntheses (Ouyang and Lee, 2014; Wang et al., 2019), and the seagrass default rests on six
715 observations. The mangrove default fares better, at 1.62 against 1.386 from 205 observations
716 (Breithaupt and Steinmuller, 2022) and 1.74 in an earlier synthesis (Alongi, 2014).

717 The direction of these disagreements is not consistent, and that is the more useful observation.
718 Irish tidal marsh accumulates at 0.29 to 0.38 t C ha^{-1} yr^{-1} (Burke et al., 2022), roughly a third
719 of the same default that the global syntheses find two to three times too low. The defect is not
720 that the central value is wrong but that a single global value is being asked to serve as a local
721 one, and its error changes sign with region. Confidence intervals do not repair this, because
722 they describe dispersion around a mean rather than a systematic regional offset. The same
723 problem appears wherever a land-use label stands in for the process that moves the carbon,
724 and Yan et al. (2018) show it outside the wetland setting, where cropland converted to poplar
725 plantation in the Ili River Valley gained soil carbon while cropland converted to spruce under
726 the same restoration programme lost it, so one conversion class contains both signs.

727 The grassland factors are the oldest in the guidance, with F_{MG} of 1.14 and 1.17 and F_I of 1.11
728 tracing to Ogle et al. (2004), a 49-study synthesis, and have passed through two revisions
729 unchanged. Measured rates cluster near 0.28 Mg C ha^{-1} yr^{-1} (Conant et al., 2017), and Wilcots
730 et al. (2022) found thirteen years of nitrogen addition leaving soil carbon stocks unchanged,
731 which questions whether the improvement factor is fertilisation-driven at all. Ireland's own Tier
732 2 land-use factors go further and contest the sign, averaging 1.05 against the default 1.14 and
733 falling to 0.85 on gley soils, roughly a fifth of the national soil-association area. On those soils
734 the default reports a gain where the national data record a loss.

735 The Irish grassland comparison of section 3.7 turns this into something the confidence
736 intervals cannot absorb. Shallow drainage lowers carbon dioxide loss and raises methane
737 simultaneously, so any single climate-zone default must be wrong on one gas whenever it is
738 right on the other. An analyst propagating the published Tier 1 intervals would conclude that

both factors are uncertain but unbiased; in Ireland both are biased, in opposite directions, and bias of this kind is invisible to the uncertainty method the guidance provides.

4.2 The nutrient fallback

Where the nutrient status of an organic soil was unknown, the Supplement assigned boreal sites the nutrient-poor factor and temperate sites the nutrient-rich one. The rule applied to drained soils in Chapter 2 (Drösler et al., 2014) and to rewetted soils in Chapter 3, which stated it for carbon dioxide and again for methane (Blain et al., 2014).

Ireland is temperate, so the rule assigns nutrient-rich factors of +0.5 t CO_2-C ha^{-1} yr^{-1} and 216 kg CH_4-C ha^{-1} yr^{-1}. Its peatlands are overwhelmingly ombrotrophic, roughly 1.4 Mha of bog against some 20,000 ha of fen, for which the physically appropriate nutrient-poor factors are −0.23 and 92. The fallback therefore converts a modelled sink into a modelled source by assumption alone, in the direction that penalises the intervention the guidance exists to encourage. The evidence beneath the split is also thin, because every Irish nutrient-rich datum comes from industrially cutaway raised bog, where extraction removed the ombrotrophic dome and exposed richer basal peat, or from a single deep-drained grassland site, and none comes from a fen. The class that governs the sign of the result is populated by sites that do not belong to it.

4.3 The salinity gate

Supplement Table 4.14 switches methane off entirely above 18 ppt (Kennedy et al., 2014), a threshold that traces to Poffenbarger et al. (2011), and while its location is broadly supported, its form is not. Salinity lowers the highest methane fluxes rather than switching methane off, salinity correlates with methane in salt marsh but not in seagrass or mangrove (Al-Haj and Fulweiler, 2020), and plant species alone explains 62 per cent of tidal marsh methane variance while salinity shows low predictive performance (Wilson et al., 2026). Since the results above

showed the threshold reversing the sign of the coastal pathway, an on-or-off switch on a weakly predictive variable bears more accounting weight than the evidence supports.

Salinity acts on methane through microbes, since the sulfate in seawater favours sulfate reduction over methanogenesis (Herbert et al., 2015), and sulfate-reducing bacteria outcompete methanogens for hydrogen and acetate (Schorn et al., 2022). Methylated compounds released by plants are used mainly by methanogens (King, 1984; Schorn et al., 2022). In sediment cores incubated from a Mediterranean seagrass meadow, these compounds sustained a median methane release of 106 μmol m^{-2} d^{-1} into the overlying water (Schorn et al., 2022). Incubated sediment from a Delaware tidal marsh creek, where June high-tide salinity averaged 30 practical salinity units, produced more methane from a methylated substrate than from ethanol (Block et al., 2025). Its authors concluded that “it is not only the presence of sulfate in polyhaline salt marshes that controls methane production but also the contribution of methyl-containing osmolytes from marsh plants and animals that support methane production in higher-salinity coastal environments” (Block et al., 2025). That marsh lay above the 18 ppt threshold at which Supplement Table 4.14 set methane to zero (Kennedy et al., 2014). In microcosm incubations of 129 sediment samples from the Chinese coast, mangrove and salt marsh sediments had a greater greenhouse gas production capacity than mudflat and seagrass sediments (Ni et al., 2025). They also held more diverse and abundant microbes (Ni et al., 2025). In field chambers on the bare tidal flats of eleven Chinese coastal marshes, methane varied with temperature, soil redox and organic carbon, and with microbial biomass and bacterial abundance (Li et al., 2020). In a Delaware salt marsh on a tidal river of about 8 to 25 ppt, porewater methane exceeded 200 μM at 50 to 75 cm, with sulfate up to 17 mM (Seyfferth et al., 2020). The study attributed the methane to methanogens using methylated compounds, and concluded that “while efflux appears to be low, it should not be assumed that these ecosystems do not produce CH_4” (Seyfferth et al., 2020). Incubations measure what sediment can produce or release rather than what reaches the atmosphere, yet both kinds of evidence link methane to the microbes present as well as to

salinity. Microbial communities differ between ecosystems and places, so a default keyed to salinity alone omits a source of variation, which widens the uncertainty of any regional estimate built on it.

For inventory analysts, the recommendation is to report each pathway's soil carbon gain beside its methane, nitrous oxide and inventory period, and to state the metric and horizon. None of this requires new data, and paragraph 37 of the Paris Agreement reporting rules already allows other metrics as supplemental information (UNFCCC, 2019).

For crediting the problem is harder, because the Wetlands Supplement supplies a flux method and no baseline-and-credit structure. Wetland Drainage and Rewetting under the Kyoto Protocol Supplement (IPCC, 2014) has no project boundary and no project baseline, because its comparison is a fixed base year under net-net accounting. A project standard that imports the Tier 1 factors without a drained baseline will credit the absolute gain, which the inland wetland mineral soil results show can be smaller than its own methane offset. This matters at the sector's current scale, given that soil carbon protocols may overestimate stocks by up to 2.5 times (Dupla et al., 2024), that Australian issuance in 2023 implied 2 to 8 t C ha^{-1} yr^{-1} against a literature range of 0.1 to 1.2 (Mitchell et al., 2024), and that blue carbon delivery stands at roughly 2 per cent of estimated potential across 81 projects (Farahmand et al., 2025).

4.4 The EU response

On 9 July 2026 the European Commission adopted the certification methodologies for carbon farming under the Union's carbon removals framework (European Commission, 2026). One of the three covers rewetting and restoration of peatlands and other organic soils, the first binding instrument to specify how a reported soil carbon removal becomes a creditable one, and its provisions run in the same direction as every argument above.

Four features matter, the first being that Tier 1 defaults are excluded from the pool this paper concerns, because the Annex permits only models and proxies for soil organic carbon in organic soils and confines defaults to ancillary terms, so the soil flux the intervention is about cannot be taken from a default table at all. The second is that methane must be quantified rather than assumed, since a proxy model must rest on a meta-analysis covering at least carbon dioxide and methane and be validated on year-round measurements at no fewer than 100 sites, with off-site fluvial losses mandatory. The third is that the methane overshoot is priced, because for the first five years either the absence of a methane peak must be made plausible from monitored vegetation and water table, or ten tonnes of carbon dioxide equivalent per hectare are deducted each year, a charge exceeding the entire GWP-100 benefit of the five weakest rows in the ledger above. The fourth is that the crediting horizon is capped by the substrate rather than by a convention, at ten to thirty years or the time to exhaust the peat at 1.0 cm per year on ombrotrophic bogs and 1.5 cm on fens, whichever is shorter, which is the first inventory period in the guidance derived from a physical quantity.

Two provisions fit less well, in that the uncertainty deduction is set at eight per cent while the recitals describe a floor of at least ten, and the two are left unreconciled. Peatland rewetting is also exempted from any liability mechanism or buffer pool on grounds of irreversibility, which the results above complicate, since a rewetted site can revert through drainage failure or drought and the near-natural reference state is itself a net source once methane is counted.

The wider point is that the instrument now governing this in Europe has reached the same conclusion this analysis reaches from the factor tables. A Tier 1 default is adequate for reporting a national aggregate and inadequate for crediting a hectare, and the gases that decide whether an intervention helps cannot be left to a default that assumes them away.

4.5 Limitations

Three limitations bound the analysis, the first being that uncertainty is propagated for the organic soil benefit alone, so the coastal, inland mineral and grassland results are point estimates, and where those sections report a sign they report it without a probability attached.

The reference stock analysis encodes the Tropical, Warm temperate and Cool temperate rows of Table 2.3 of the 2019 Refinement, but not the Polar and Boreal rows. The skew of the reference stocks is therefore measured across eight of the ten published climate zones. The organic soil benefit is unaffected, since it takes its factors from the Wetlands Supplement rather than from Table 2.3.

The third concerns the Irish case, which rests on three independent evidence streams. These are published emission factors needing no data at all, the national soils map, and Ireland's own reporting tables.

The coastal arm is the weakest, because the saltmarsh-to-salinity join classified only a tenth of the resource, Ireland having measured salinity in nine of its 195 transitional water bodies, and it uses water-column values as a proxy for marsh porewater, so it bounds the exposure rather than measuring it. Nothing in the Irish case rests on seagrass, for which no measurement exists anywhere on the island.

5 Conclusions

A reported soil carbon removal is not a climate benefit, and the guidance that permits the removal supplies no way to tell the two apart. Assembling the Tier 1 and Tier 2 removal pathways on a single greenhouse-gas-weighted axis shows how far apart the two can lie. Rewetting organic soils is beneficial at the published point estimates in every Tier 1 default combination at GWP-100 but harmful in five of thirteen at GWP-20, and the probability of benefit falls below 0.95 in the same five at GWP-100 alone. Inland wetland mineral soil

rewetting reports a positive stock change that its own methane factor exceeds in all nine climate regions for twenty years, with no drained baseline in Chapter 5 to offset against. A restored tidal marsh below 18 ppt is a net source in absolute terms while remaining a benefit of some 27 t CO_2e ha^{-1} yr^{-1} against the drained state, and fertilised improved grassland crosses into net source because a twenty-year gain is set against a perpetual penalty. In Ireland's reported inventory the same pathway is worth about seven tonnes of carbon dioxide equivalent per hectare per year at GWP-100 and costs nearly three at GWP-20, with no factor changing between the two lines. The defaults beneath those pathways are the second gap, and the published uncertainty range discloses neither. The reference stocks behind every mineral soil removal exceed their source median in 34 of 37 cells, by 13.0 per cent at the median and 154.5 per cent at most. That excess does not show a national total to be inflated, but the typical parcel holds less carbon than its default. The error printed beside each stock is the precision of a global mean, which replaced a dispersion between the 2006 and 2019 tables without notice. It understates the range a single parcel faces about eight-fold.

The common structure is that each removal is reported apart from its own offsets, and the two are added only in a national total weighted at a single 100-year horizon. The remedy does not require new science, but only that the offsetting fluxes, the inventory period, and the emission metric with its horizon be reported alongside the gain, and that any pathway whose sign moves across a defensible set of metrics be reported as metric-dependent rather than as a benefit.

That remedy is no longer hypothetical, because in adopting its carbon farming certification methodologies in July 2026 the European Union excluded default emission factors from the organic soil pool outright, required methane to be quantified rather than assumed, priced the post-rewetting methane transient absent evidence to the contrary, and capped the crediting horizon at the physical exhaustion of the peat (European Commission, 2026). The distinction the regulation draws is the one this paper argues for, namely that a Tier 1 default is a reasonable basis for reporting a national aggregate and an inadequate one for crediting a

hectare, because reporting tolerates errors that cancel across a country while crediting does not.

Declarations

Ethics approval and consent to participate

Not applicable. This study analyses published emission factors and public datasets and involves no human participants, human data or animals.

Consent for publication

Not applicable.

Data availability statement

All emission factors and reference stocks used here are transcribed from the published IPCC guidance and are encoded in the manuscript source, so the analysis reproduces from the document itself. The four external datasets, their provenance and the derivation scripts that process them are described in section 2.8. The Irish Soil Information System national soils map, the NPWS Saltmarsh Monitoring Project layer and the EPA Water Framework Directive water bodies are third-party datasets available from their respective public services under their own terms. The manuscript source, the derivation scripts, the derived tables and the figure code are archived at https://github.com/seamusrobertmurphy/ipcc-carbon-gains.

Competing interests

The author declares no competing interests.

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Author contributions

Dr Seamus Murphy: conceptualization; data curation; formal analysis; investigation; methodology; software; validation; visualization; writing, original draft; writing, review and editing.

Use of generative artificial intelligence

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References

Aitova, E., Morley, T., Wilson, D., Renou-Wilson, F., 2023. A review of greenhouse gas emissions and removals from Irish peatlands. Mires and Peat 29, 04. *https://doi.org/10.19189/map.2022.snpg.sta.2414*

Al-Haj, A.N., Fulweiler, R.W., 2020. A synthesis of methane emissions from shallow vegetated coastal ecosystems. Global Change Biology 26, 2988–3005. *https://doi.org/10.1111/gcb.15046*

930 Allen, M.R., Peters, G.P., Shine, K.P., Azar, C., Balcombe, P., Boucher, O., Cain, M., Ciais,
931 P., Collins, W., Forster, P.M., Frame, D.J., Friedlingstein, P., Fyson, C., Gasser, T.,
932 Hare, B., Jenkins, S., Hamburg, S.P., Johansson, D.J.A., Lynch, J., Macey, A.,
933 Morfeldt, J., Nauels, A., Ocko, I., Oppenheimer, M., Pacala, S.W., Pierrehumbert, R.,
934 Rogelj, J., Schaeffer, M., Schleussner, C.F., Shindell, D., Skeie, R.B., Smith, S.M.,
935 Tanaka, K., 2022. Indicate separate contributions of long-lived and short-lived
936 greenhouse gases in emission targets. npj Climate and Atmospheric Science 5.
937 *https://doi.org/10.1038/s41612-021-00226-2*

938 Alongi, D.M., 2014. Carbon cycling and storage in mangrove forests. Annual Review of Marine
939 Science 6, 195–219. *https://doi.org/10.1146/annurev-marine-010213-135020*

940 Batjes, N.H., 2011. Soil organic carbon stocks under native vegetation – revised estimates for
941 use with the simple assessment option of the carbon benefits project system.
942 Agriculture, Ecosystems & Environment 142, 365–373.
943 *https://doi.org/10.1016/j.agee.2011.06.007*

944 Batjes, N.H., 2010. A global framework of soil organic carbon stocks under native vegetation
945 for use with the simple assessment option of the carbon benefits project system (ver.
946 1.0) (ISRIC Report No. 2010/10). ISRIC World Soil Information, Wageningen, the
947 Netherlands.

948 Blain, D., Murdiyarso, D., Couwenberg, J., Nagata, O., Renou-Wilson, F., Sirin, A., Strack, M.,
949 Tuittila, E.-S., Wilson, D., 2014. Chapter 3: Rewetted organic soils, in: Hiraishi, T.,
950 Krug, T., Tanabe, K., Srivastava, N., Baasansuren, J., Fukuda, M., Troxler, T.G. (Eds.),
951 2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas
952 Inventories: Wetlands. IPCC, Switzerland.

953 Block, K.R., Arbetman, A., Slotznick, S.P., Hanson, T.E., Luther, G.W., III, Shah Walter, S.R.,
954 2025. Influence of carbon source and iron oxide minerals on methane production and

magnetic mineral formation in salt marsh sediments. Biogeosciences 22, 5991–6015. https://doi.org/10.5194/bg-22-5991-2025

Breithaupt, J.L., Steinmuller, H.E., 2022. Refining the global estimate of mangrove carbon burial rates using sedimentary and geomorphic settings. Geophysical Research Letters 49. https://doi.org/10.1029/2022gl100177

Burke, S.A., Manahan, J., Eichelmann, E., Cott, G.M., 2022. Dublin's saltmarshes contain climate-relevant carbon pools. Frontiers in Marine Science 9, 976457. https://doi.org/10.3389/fmars.2022.976457

Conant, R.T., Cerri, C.E.P., Osborne, B.B., Paustian, K., 2017. Grassland management impacts on soil carbon stocks: A new synthesis. Ecological Applications 27, 662–668. https://doi.org/10.1002/eap.1473

Domke, G., Brandon, A., Diaz-Lasco, R., Federici, S., Garcia-Apaza, E., Grassi, G., Gschwantner, T., Herold, M., Hirata, Y., Kasimir, Å., Kinyanjui, M.J., Krisnawati, H., Lehtonen, A., Malimbwi, R.E., Niinistö, S., Ogle, S.M., Paul, T., Ravindranath, N.H., Rock, J., Sanquetta, C.R., Sanz Sanchez, M.J., Vitullo, M., Wakelin, S.J., Zhu, J., 2019. Chapter 4: Forest land, in: Calvo Buendia, E., Tanabe, K., Kranjc, A., Baasansuren, J., Fukuda, M., Ngarize, S., Osako, A., Pyrozhenko, Y., Shermanau, P., Federici, S. (Eds.), 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4: Agriculture, Forestry and Other Land Use. IPCC, Switzerland.

Drösler, M., Verchot, L.V., Freibauer, A., Pan, G., Evans, C.D., Bourbonniere, R.A., Alm, J.P., Page, S., Agus, F., Hergoualc'h, K., Couwenberg, J., Jauhiainen, J., Sabiham, S., Wang, C., 2014. Chapter 2: Drained inland organic soils, in: Hiraishi, T., Krug, T., Tanabe, K., Srivastava, N., Baasansuren, J., Fukuda, M., Troxler, T.G. (Eds.), 2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands. IPCC, Switzerland.

Dupla, X., Bonvin, E., Deluz, C., Lugassy, L., Verrecchia, E., Baveye, P.C., Grand, S., Boivin, P., 2024. Are soil carbon credits empty promises? Shortcomings of current soil carbon quantification methodologies and improvement avenues. Soil Use and Management 40. *https://doi.org/10.1111/sum.13092*

European Commission, 2026. Commission delegated regulation supplementing regulation (EU) 2024/3012 of the European parliament and of the council by establishing the certification methodologies for carbon farming activities.

Evans, C.D., Peacock, M., Baird, A.J., Artz, R.R.E., Burden, A., Callaghan, N., Chapman, P.J., Cooper, H.M., Coyle, M., Craig, E., Cumming, A., Dixon, S., Gauci, V., Grayson, R.P., Helfter, C., Heppell, C.M., Holden, J., Jones, D.L., Kaduk, J., Levy, P., Matthews, R., McNamara, N.P., Misselbrook, T., Oakley, S., Page, S.E., Rayment, M., Ridley, L.M., Stanley, K.M., Williamson, J.L., Worrall, F., Morrison, R., 2021. Overriding water table control on managed peatland greenhouse gas emissions. Nature 593, 548–552. *https://doi.org/10.1038/s41586-021-03523-1*

Farahmand, S., Hilmi, N., Duarte, C.M., 2025. The rise and flows of blue carbon credits advance global climate and biodiversity goals. npj Ocean Sustainability 4. *https://doi.org/10.1038/s44183-025-00141-6*

Forster, P., Storelvmo, T., Armour, K., Collins, W., Dufresne, J.-L., Frame, D., Lunt, D.J., Mauritsen, T., Palmer, M.D., Watanabe, M., Wild, M., Zhang, H., 2021. The earth's energy budget, climate feedbacks and climate sensitivity, in: Climate Change 2021: The Physical Science Basis. Contribution of Working Group i to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press, Cambridge, United Kingdom; New York, NY, USA, pp. 923–1054. *https://doi.org/10.1017/9781009157896.009*

1005 Gao, J., Feng, J., Zhang, X., Yu, F.-H., Xu, X., Kuzyakov, Y., 2016. Drying-rewetting cycles
1006 alter carbon and nitrogen mineralization in litter-amended alpine wetland soil. Catena
1007 145, 285–290. *https://doi.org/10.1016/j.catena.2016.06.026*

1008 Gao, J.-Q., Duan, M.-Y., Zhang, X.-Y., Li, Q.-W., Yu, F.-H., 2018. Effects of frequency and
1009 intensity of drying-rewetting cycles on Hydrocotyle vulgaris growth and greenhouse
1010 gas emissions from wetland microcosms. Catena 164, 44–49.
1011 *https://doi.org/10.1016/j.catena.2018.01.006*

1012 Günther, A., Barthelmes, A., Huth, V., Joosten, H., Jurasinski, G., Koebsch, F., Couwenberg,
1013 J., 2020. Prompt rewetting of drained peatlands reduces climate warming despite
1014 methane emissions. Nature Communications 11. *https://doi.org/10.1038/s41467-020-*
1015 *15499-z*

1016 He, H., Roulet, N.T., 2023. Improved estimates of carbon dioxide emissions from drained
1017 peatlands support a reduction in emission factor. Communications Earth &
1018 Environment 4. *https://doi.org/10.1038/s43247-023-01091-y*

1019 Herbert, E.R., Boon, P., Burgin, A.J., Neubauer, S.C., Franklin, R.B., Ardón, M.,
1020 Hopfensperger, K.N., Lamers, L.P.M., Gell, P., 2015. A global perspective on wetland
1021 salinization: Ecological consequences of a growing threat to freshwater wetlands.
1022 Ecosphere 6, 1–43. *https://doi.org/10.1890/ES14-00534.1*

1023 IPCC, 2019. 2019 refinement to the 2006 IPCC guidelines for national greenhouse gas
1024 inventories. Calvo Buendia, E., Tanabe, K., Kranjc, A., Baasansuren, J., Fukuda, M.,
1025 Ngarize, S., Osako, A., Pyrozhenko, Y., Shermanau, P.; Federici, S. (eds). IPCC,
1026 Switzerland.

1027 IPCC, 2014. 2013 revised supplementary methods and good practice guidance arising from
1028 the kyoto protocol. Hiraishi, T., Krug, T., Tanabe, K., Srivastava, N., Baasansuren, J.,
1029 Fukuda, M.; Troxler, T.G. (eds). IPCC, Switzerland.

IPCC, 2006. 2006 IPCC guidelines for national greenhouse gas inventories, volume 4: Agriculture, forestry and other land use. Prepared by the National Greenhouse Gas Inventories Programme. Eggleston, H.S., Buendia, L., Miwa, K., Ngara, T.; Tanabe, K. (eds). IGES, Japan.

Kalhori, A., Wille, C., Gottschalk, P., Li, Z., Hashemi, J., Kemper, K., Sachs, T., 2024. Temporally dynamic carbon dioxide and methane emission factors for rewetted peatlands. Communications Earth & Environment 5. *https://doi.org/10.1038/s43247-024-01226-9*

Kennedy, H., Alongi, D.M., Karim, A., Chen, G., Chmura, G.L., Crooks, S., Kairo, J.G., Liao, B., Lin, G., 2014. Chapter 4: Coastal wetlands, in: Hiraishi, T., Krug, T., Tanabe, K., Srivastava, N., Baasansuren, J., Fukuda, M., Troxler, T.G. (Eds.), 2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands. IPCC, Switzerland.

King, G.M., 1984. Metabolism of trimethylamine, choline, and glycine betaine by sulfate-reducing and methanogenic bacteria in marine sediments. Applied and Environmental Microbiology 48, 719–725. *https://doi.org/10.1128/aem.48.4.719-725.1984*

Launiainen, S., Ahtikoski, A., Rinne, J., Ojanen, P., Hökkä, H., 2025. Rewetting drained boreal peatland forests does not mitigate climate warming in the twenty-first century. Ambio 54, 2107–2117. *https://doi.org/10.1007/s13280-025-02225-6*

Li, C., Frolking, S., Butterbach-Bahl, K., 2005. Carbon sequestration in arable soils is likely to increase nitrous oxide emissions, offsetting reductions in climate radiative forcing. Climatic Change 72, 321–338. *https://doi.org/10.1007/s10584-005-6791-5*

Li, X., Sardans, J., Hou, L., Liu, M., Xu, C., Peñuelas, J., 2020. Climatic temperature controls the geographical patterns of coastal marshes greenhouse gases emissions over China. Journal of Hydrology 590, 125378. *https://doi.org/10.1016/j.jhydrol.2020.125378*

Lovelock, C.E., Evans, C., Barros, N., Prairie, Y., Alm, J., Bastviken, D., Beaulieu, J.J., Garneau, M., Harby, A., Harrison, J., Pare, D., Raadal, H.L., Sherman, B., Zhang, C., Ogle, S.M., 2019. Chapter 7: wetlands, in: Calvo Buendia, E., Tanabe, K., Kranjc, A., Baasansuren, J., Fukuda, M., Ngarize, S., Osako, A., Pyrozhenko, Y., Shermanau, P., Federici, S. (Eds.), 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4: Agriculture, Forestry and Other Land Use. IPCC, Switzerland.

Lugato, E., Leip, A., Jones, A., 2018. Mitigation potential of soil carbon management overestimated by neglecting N2O emissions. Nature Climate Change 8, 219–223. *https://doi.org/10.1038/s41558-018-0087-z*

Ma, S., Creed, I.F., Badiou, P., 2024. New perspectives on temperate inland wetlands as natural climate solutions under different CO2-equivalent metrics. npj Climate and Atmospheric Science 7. *https://doi.org/10.1038/s41612-024-00778-z*

McConkey, B., Ogle, S.M., Chirinda, N., Kishimoto-Mo, A.W., Baldock, J., Trunov, A., 2019. Chapter 6: grassland, in: Calvo Buendia, E., Tanabe, K., Kranjc, A., Baasansuren, J., Fukuda, M., Ngarize, S., Osako, A., Pyrozhenko, Y., Shermanau, P., Federici, S. (Eds.), 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4: Agriculture, Forestry and Other Land Use. IPCC, Switzerland.

Mitchell, E., Takeda, N., Grace, L., Grace, P., Day, K., Ahmadi, S., Badgery, W., Cowie, A., Simmons, A., Eckard, R., Harrison, M.T., Parton, W., Wilson, B., Orgill, S., Viscarra Rossel, R.A., Pannell, D., Stanley, P., Deane, F., Rowlings, D., 2024. Making soil carbon credits work for climate change mitigation. Carbon Management 15. *https://doi.org/10.1080/17583004.2024.2430780*

Ni, L., Fu, C., Li, G., Feng, K., Deng, Y., Wu, J., Duarte, C.M., Li, Y., Zhang, D., 2025. Microbial abundance shapes latitudinal pattern of the greenhouse gas production potential in

blue carbon ecosystems. Water Research 287, 124357. https://doi.org/10.1016/j.watres.2025.124357

Ogle, S.M., Conant, R.T., Paustian, K., 2004. Deriving grassland management factors for a carbon accounting method developed by the intergovernmental panel on climate change. Environmental Management 33. https://doi.org/10.1007/s00267-003-9105-6

Ogle, S.M., Kurz, W.A., Green, C., Brandon, A., Baldock, J., Domke, G., Herold, M., Bernoux, M., Chirinda, N., Ligt, R. de, Federici, S., Garcia-Apaza, E., Grassi, G., Gschwantner, T., Hirata, Y., Houghton, R., House, J.I., Ishizuka, S., Jonckheere, I., Krisnawati, H., Lehtonen, A., Kinyanjui, M.J., McConkey, B., Næsset, E., Niinistö, S.M., Ometto, J.P., Panichelli, L., Paul, T., Petersson, H., Reddy, S., Regina, K., Rocha, M.T., Rock, J., Sanz Sanchez, M.J., Sanquetta, C., Sato, A., Somogyi, Z., Trunov, A., Vazquez-Amabile, G., Vitullo, M., Wang, C., Waterworth, R.M., 2019. Chapter 2: Generic methodologies applicable to multiple land-use categories, in: Calvo Buendia, E., Tanabe, K., Kranjc, A., Baasansuren, J., Fukuda, M., Ngarize, S., Osako, A., Pyrozhenko, Y., Shermanau, P., Federici, S. (Eds.), 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4: Agriculture, Forestry and Other Land Use. IPCC, Switzerland.

Ojanen, P., Minkkinen, K., 2020. Rewetting offers rapid climate benefits for tropical and agricultural peatlands but not for forestry-drained peatlands. Global Biogeochemical Cycles 34. https://doi.org/10.1029/2019gb006503

Ouyang, X., Lee, S.Y., 2014. Updated estimates of carbon accumulation rates in coastal marsh sediments. Biogeosciences 11, 5057–5071. https://doi.org/10.5194/bg-11-5057-2014

Petersson, T., Antoniella, G., Chiriacò, M.V., Perugini, L., Chiti, T., 2024. The misconception of soil organic carbon sequestration notion: When do we achieve climate benefit? Soil Use and Management 40, e13009. https://doi.org/10.1111/sum.13009

Phillips, R., Beeri, O., 2008. The role of hydropedologic vegetation zones in greenhouse gas emissions for agricultural wetland landscapes. Catena 72, 386–394. https://doi.org/10.1016/j.catena.2007.07.007

Poffenbarger, H.J., Needelman, B.A., Megonigal, J.P., 2011. Salinity influence on methane emissions from tidal marshes. Wetlands 31, 831–842. https://doi.org/10.1007/s13157-011-0197-0

Rahman, N., Richards, K.G., Harty, M.A., Watson, C.J., Carolan, R., Krol, D., Lanigan, G.J., Forrestal, P.J., 2021. Differing effects of increasing calcium ammonium nitrate, urea and urea + NBPT fertiliser rates on nitrous oxide emission factors at six temperate grassland sites in Ireland. Agriculture, Ecosystems & Environment 313, 107382. https://doi.org/10.1016/j.agee.2021.107382

Schorn, S., Ahmerkamp, S., Bullock, E., Weber, M., Lott, C., Liebeke, M., Lavik, G., Kuypers, M.M.M., Graf, J.S., Milucka, J., 2022. Diverse methylotrophic methanogenic archaea cause high methane emissions from seagrass meadows. Proceedings of the National Academy of Sciences 119, e2106628119. https://doi.org/10.1073/pnas.2106628119

Schuster, L., Taillardat, P., Macreadie, P.I., Malerba, M.E., 2024. Freshwater wetland restoration and conservation are long-term natural climate solutions. Science of The Total Environment 922, 171218. https://doi.org/10.1016/j.scitotenv.2024.171218

Schwalm, M., Zeitz, J., 2015. Concentrations of dissolved organic carbon in peat soils as influenced by land use and site characteristics, a lysimeter study. Catena 127, 72–79. https://doi.org/10.1016/j.catena.2014.12.007

Seyfferth, A.L., Bothfeld, F., Vargas, R., Stuckey, J.W., Wang, J., Kearns, K., Michael, H.A., Guimond, J., Yu, X., Sparks, D.L., 2020. Spatial and temporal heterogeneity of geochemical controls on carbon cycling in a tidal salt marsh. Geochimica et Cosmochimica Acta 282, 1–18. https://doi.org/10.1016/j.gca.2020.05.013

UNFCCC, 2019. *Decision 18/CMA.1. Modalities, procedures and guidelines for the transparency framework for action and support referred to in Article 13 of the Paris Agreement*.

Vanselow-Algan, M., Schmidt, S.R., Greven, M., Fiencke, C., Kutzbach, L., Pfeiffer, E.-M., 2015. High methane emissions dominated annual greenhouse gas balances 30 years after bog rewetting. Biogeosciences 12, 4361–4371. *https://doi.org/10.5194/bg-12-4361-2015*

Villarino, S.H., Studdert, G.A., Laterra, P., 2018. Greenhouse gas inventories: Deriving soil organic carbon change factors and assessing soil depth relevance in Argentinean Semiarid Chaco. Catena 169, 164–174. *https://doi.org/10.1016/j.catena.2018.05.041*

Wang, F., Lu, X., Sanders, C.J., Tang, J., 2019. Tidal wetland resilience to sea level rise increases their carbon sequestration capacity in United States. Nature Communications 10. *https://doi.org/10.1038/s41467-019-13294-z*

Wickland, K.P., Krusche, A.V., Kolka, R.K., Kishimoto-Mo, A.W., Chimner, R.A., Serengil, Y., 2014. Chapter 5: Inland wetland mineral soils, in: Hiraishi, T., Krug, T., Tanabe, K., Srivastava, N., Baasansuren, J., Fukuda, M., Troxler, T.G. (Eds.), 2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands. IPCC, Switzerland.

Wilcots, M.E., Schroeder, K.M., DeLancey, L.C., Kjaer, S.J., Hobbie, S.E., Seabloom, E.W., Borer, E.T., 2022. Realistic rates of nitrogen addition increase carbon flux rates but do not change soil carbon stocks in a temperate grassland. Global Change Biology 28, 4819–4831. *https://doi.org/10.1111/gcb.16272*

Wilson, D., Blain, D., Couwenberg, J., Evans, C.D., Murdiyarso, D., Page, S.E., Renou-Wilson, F., Rieley, J.O., Sirin, A., Strack, M., Tuittila, E.-S., 2016. Greenhouse gas emission factors associated with rewetting of organic soils. Mires and Peat 17, 04. *https://doi.org/10.19189/map.2016.omb.222*

Wilson, E.M., Balint, S.J., Fulweiler, R.W., 2026. Plants are a powerful proxy for global tidal marsh methane fluxes. Proceedings of the National Academy of Sciences 123, e2530848123. *https://doi.org/10.1073/pnas.2530848123*

Yan, M., Zhang, W., Zhang, Z., Wang, L., Ren, H., Jiang, Y., Zhang, X., 2018. Responses of soil C stock and soil C loss to land restoration in Ili River Valley, China. Catena 171, 469–474. *https://doi.org/10.1016/j.catena.2018.07.035*

Zhang, X., Ots, M., Johnston, E., Brown, K., Doar, N., Lynch, J., 2025. Peatland restoration is anticipated to provide climate change mitigation over all time-scales: A UK case-study. Environmental Research Communications 7, 115011. *https://doi.org/10.1088/2515-7620/ae14bb*

Supplementary material

The tables below give the model inputs, sensitivity analyses, supporting result matrices and scale calculations on which the main text draws.

Table S1. AR6 Table 7.15 emission metrics used throughout. Biogenic methane is weighted at 27.0 at GWP-100, not at a value adjusted for the molar mass of carbon.

gas	gwp20	gwp100	gtp100
CH4_fossil	82.5	29.8	7.5
CH4_nonfossil	79.7	27.0	4.7
N2O	273.0	273.0	233.0

Table S2. Model input: rewetted organic soils, t CO_2e ha^{-1} yr^{-1}, positive = source. CO_2 only is the reported soil carbon term; the net column adds DOC and methane. These are absolute fluxes, not benefits.

Zone	Nutrient status	n	CO_2 only	Absolute net GWP-100
Boreal	Poor	26	-1.25	0.52
Boreal	Rich	39	-2.02	3.21
Temperate	Poor	43	-0.84	3.35
Temperate	Rich	15	1.83	10.49
Tropical	All	11	0.00	3.35

Table S3. Benefit of rewetting, t CO_2e ha^{-1} yr^{-1}. Positive means rewetting reduces net emissions. Drained net flux and rewetted net flux are at GWP-100, positive meaning a net emission. Drained includes CO_2, DOC, land and ditch methane weighted by Equation 2.6, and direct nitrous oxide; rewetted includes CO_2, DOC and methane, with nitrous oxide assumed negligible at Tier 1. Indirect nitrous oxide is omitted on the drained side, which understates the benefit.

Land use	Zone	Drained nutrient	Rewetted nutrient	n	Drained net flux	Rewetted net flux	GWP-100	GWP-20
Forest Land	Boreal	Poor	Poor	59	1.78	0.52	1.26	-0.97
Forest Land	Boreal	Rich	Rich	62	5.42	3.21	2.21	-7.03
Forest Land	Temperate	Unsplit	Rich	8	12.08	10.49	1.59	-13.17
Forest Land	Tropical	Unsplit	All	21	24.82	3.35	21.47	21.22
Cropland	Temperate	Rich	Rich	39	37.25	10.49	26.76	14.65
Cropland	Tropical	Unsplit	All	10	57.89	3.35	54.54	54.40
Grassland	Boreal	Unsplit	Poor	8	27.02	0.52	26.50	26.76
Grassland	Temperate	Poor	Poor	7	24.03	3.35	20.68	17.38
Grassland (deep-drained)	Temperate	Rich	Rich	39	29.00	10.49	18.51	7.21
Grassland (shallow-drained)	Temperate	Rich	Rich	13	16.73	10.49	6.24	-5.59
Grassland	Tropical	Unsplit	All	NA	41.76	3.35	38.41	38.27
Peat extraction	Temperate	Unsplit	Rich	20	12.42	10.49	1.93	-11.52
Plantation oil palm	Tropical	Unsplit	All	10	45.07	3.35	41.72	41.23

Table S4. Effect of treating the published central value as the mode of a triangular distribution on right-skewed bounds, rewetted organic soils, kg CH_4-C ha^{-1} yr^{-1}.

Zone	Nutrient	Published CH_4	Sampled mean	Ratio
Boreal	Poor	41	95.4	2.33
Boreal	Rich	137	209.5	1.55
Temperate	Poor	92	180.8	1.96
Temperate	Rich	216	359.6	1.65
Tropical	All	41	60.9	1.47

Table S5. Sensitivity of the headline count to the distributional assumption, out of thirteen rows, on identical published bounds.

Sampling distribution	Rows below P 0.95
Triangular (mode at published value)	5
Normal (mean at published value)	5
Uniform on the published bounds	6

Table S6. Coastal wetland soil carbon, t CO_2 ha^{-1} yr^{-1} removed. Headroom depends on the stock present at rewetting; the bare-start column is an upper bound, not an expectation. Table 4.11 is titled for extraction activities and its column is the stock destroyed by excavation, not a documented accumulation ceiling. Salinity columns are NA for seagrass because Table 4.14 covers tidal marsh and mangrove only.

System	n	Soil gain	Headroom	at 25% stock	at 50% stock	Net >18 ppt	Net <18 ppt
Mangrove	69	5.94	238	179	119	5.94	0.71
Tidal marsh	66	3.34	280	210	140	3.34	-1.89
Seagrass	6	1.58	NA	NA	NA	NA	NA

Table S7. Coastal wetlands with and without the drained baseline, t CO_2e ha^{-1} yr^{-1}. The absolute columns are the restored state alone, positive = removal. The benefit columns subtract the restored state from the Table 4.13 drained baseline of 28.97 t CO_2 ha^{-1} yr^{-1} and are the quantity the hypothesis concerns.

System	Absolute >18 ppt	Absolute <18 ppt	Benefit >18 ppt	Benefit <18 ppt
Mangrove	5.94	0.71	34.91	29.68
Tidal marsh	3.34	-1.89	32.31	27.08

Table S8. Inland wetland mineral soils, rewetting, t CO_2e ha^{-1} yr^{-1} at GWP-100. Positive is a removal; the methane column is an emission and is subtracted. Table 5.3 sets two consecutive 20-year blocks, so the annual gain is not constant. Methane is the zone-matched Table 5.4 factor.

Zone	SOCref	Gain yr 1-20	Gain yr 21-40	CH_4 offset	Net yr 1-20	Net yr 21-40
Boreal	116	1.91	4.25	2.05	-0.14	2.20
Cold temp dry	87	1.44	3.19	6.34	-4.90	-3.15
Cold temp moist	128	2.11	4.69	6.34	-4.23	-1.65
Warm temp dry	74	1.22	2.71	6.34	-5.12	-3.63
Warm temp moist	135	2.23	4.95	6.34	-4.11	-1.39
Tropical dry	22	0.36	0.81	24.30	-23.94	-23.49
Tropical moist	68	1.12	2.49	24.30	-23.18	-21.81
Tropical wet	49	0.81	1.80	24.30	-23.49	-22.50
Tropical montane	82	1.35	3.01	24.30	-22.95	-21.29

Table S9. Improved grassland with high input. Gains in t CO_2 ha^{-1} yr^{-1} removed, accruing over 20 years and zero thereafter.

Climate zone	Soil	SOCref (t C ha^{-1})	F after	Gain from nominal	Gain from degraded
Cool temp moist	HAC	81	1.265	3.94	8.40
Cool temp moist	LAC	76	1.265	3.70	7.88
Cool temp moist	VOL	136	1.265	6.62	14.10
Cool temp moist	WET	128	1.265	6.23	13.27
Cool temp dry	HAC	43	1.265	2.09	4.46
Cool temp dry	LAC	33	1.265	1.61	3.42
Cool temp dry	VOL	20	1.265	0.97	2.07
Cool temp dry	WET	87	1.265	4.23	9.02
Tropical montane	HAC	51	1.288	2.69	5.49
Tropical montane	LAC	44	1.288	2.32	4.74
Tropical montane	VOL	96	1.288	5.06	10.34
Tropical montane	WET	82	1.288	4.32	8.83
Tropical wet	HAC	60	1.299	3.29	6.59
Tropical wet	LAC	52	1.299	2.85	5.71
Tropical wet	VOL	77	1.299	4.22	8.45
Tropical wet	WET	49	1.299	2.68	5.38
Tropical moist	HAC	40	1.299	2.19	4.39

Climate zone	Soil	SOCref (t C ha^{-1})	F after	Gain from nominal	Gain from degraded
Tropical moist	LAC	38	1.299	2.08	4.17
Tropical moist	VOL	70	1.299	3.83	7.68
Tropical moist	WET	68	1.299	3.72	7.46
Tropical dry	HAC	21	1.299	1.15	2.30
Tropical dry	LAC	19	1.299	1.04	2.09
Tropical dry	VOL	50	1.299	2.74	5.49
Tropical dry	WET	22	1.299	1.20	2.41
Warm temp moist	HAC	64	1.265	3.11	6.63
Warm temp moist	LAC	55	1.265	2.68	5.70
Warm temp moist	VOL	138	1.265	6.71	14.30
Warm temp moist	WET	135	1.265	6.57	13.99
Warm temp dry	HAC	24	1.265	1.17	2.49
Warm temp dry	LAC	19	1.265	0.92	1.97
Warm temp dry	VOL	84	1.265	4.09	8.71
Warm temp dry	WET	74	1.265	3.60	7.67

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Table S10. The ten Table 2.3 cells in which the published mean most exceeds the median of the same profiles. Mean above median is that excess as a percentage of the median. The coefficient of variation is the spread of the profiles, and the printed error is the precision of the mean, which is what Table 2.3 reports.

Zone	Soil	Profiles	Mean (t C ha^{-1})	Median (t C ha^{-1})	Mean above median (%)	Coefficient of variation (%)	Printed error (%)
W2	VOL	10	84	33	154.5	104.8	64.9
W1	WET	28	135	94	43.6	74.8	27.7
T1	WET	12	82	58	41.4	89.0	50.4
T1	VOL	10	96	74	29.7	50.0	31.0
T3	WET	55	68	53	28.3	66.2	17.5
W2	HAC	781	24	19	26.3	66.7	4.7
W1	SAN	39	36	29	24.1	72.2	22.7
T4	HAC	554	21	17	23.5	61.9	5.2
T1	LAC	84	44	36	22.2	50.0	10.7
C1	SAN	126	51	42	21.4	76.5	13.4

Table S11. The eight largest climate-by-soil combinations by mapped global area, with the percentage by which the Table 2.3 published mean exceeds the robust median. Areas are approximate: FAO ecological zones are used as a proxy for the IPCC climate zones, which are defined climatically rather than ecologically.

Zone	Soil	Area (Mha)	Mean above median (%)	Profiles
T4	HAC	1752.3	23.5	554
C1	HAC	1124.2	9.5	334
C2	HAC	953.4	13.2	177
W2	HAC	953.4	26.3	781
T2	LAC	945.9	13.0	271
W1	HAC	668.9	8.5	489
T3	LAC	599.9	15.2	326
T4	SAN	500.7	12.5	164

Every total in Tables S12 and S13 is a rate multiplied by an area, and it inherits every weakness of both. The rates carry the metric dependence and the sampling uncertainty established in the main text. The areas carry a peatland undercount of unknown size, an unresolvable drained fraction, and for grassland a denominator that is a proxy rather than a measurement. These figures are reported to set scale, not to be cited as estimates of the global carbon flux from these pathways.

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Organic soil (Histosols): 131 Mha total; Boreal 104, Temperate 22, Tropical 5.

Wetland mineral (Gleysols): 398 Mha total.
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Table S12. Organic soil rewetting, Mt CO_2e yr^{-1}. Benefit columns are t CO_2e ha^{-1} yr^{-1} averaged over the land uses in each zone. The total columns are an upper bound on the opportunity, not a projection: they assume every mapped hectare is drained and then rewetted.

Zone	Area (Mha)	Benefit GWP-100	Benefit GWP-20	Drain-rewet GWP-100	Drain-rewet GWP-20	10% drained, GWP-100
Boreal	103.9	9.99	6.25	1038.0	649.4	103.8
Temperate	22.4	12.62	1.49	282.7	33.4	28.3
Tropical	5.0	39.03	38.78	195.2	193.9	19.5

Upper bound, all organic soil: 1516 Mt CO_2e yr^{-1} at GWP-100, 877 at GWP-20.

At an illustrative 10% drained: 152 Mt CO_2e yr^{-1} at GWP-100.

Table S13. The two pathways whose eligible area cannot be derived from soil class. No total at 100% is given: Chapter 5 applies only to cultivated wetland mineral soil being rewetted, and the grassland denominator is ecological zone rather than managed pasture. The columns show what the rate implies at illustrative eligible fractions, which is the most that can be said.

Pool & activity pathway	Area (Mha)	t CO_2e ha^{-1} yr^{-1}	Mt CO_2e yr^{-1} at 5% eligible	Mt CO_2e yr^{-1} at 10% eligible	Mt CO_2e yr^{-1} at 25% eligible
Wetland mineral soil, rewetting	398	-12.45	-248	-496	-1239
Improved grassland	1830	1.30	119	238	595

Wetland mineral soil: 398 Mha mapped, -12.45 t CO_2e ha^{-1} yr^{-1}, a net source per hectare.

At 10% cultivated and rewetted that is -495 Mt CO_2e yr^{-1}.

Grassland proxy: 1830 Mha, 1.30 t CO_2 ha^{-1} yr^{-1}; at 10% improved, 238 Mt CO_2 yr^{-1} before any N_2O.

Table S14. Irish country-specific factors against the Wetlands Supplement temperate defaults, after Aitova et al. (2023) Table 2. CO_2 in t C ha^{-1} yr^{-1}, CH_4 in kg CH_4-C ha^{-1} yr^{-1}. A ratio below 1 means the Irish measurement is lower than the default.

Land use	Nutrient	CO_2 Tier 1	CO_2 Irish	Ratio	CH_4 Tier 1	CH_4 Irish	Ratio
Grassland	Poor	5.30	1.30	0.25	1.4	8.82	6.30
Grassland deep-drained	Rich	6.10	5.08	0.83	12.0	-0.75	-0.06
Industrial cutaway	Poor	2.80	1.21	0.43	4.6	0.00	0.00
Industrial cutaway	Rich	2.80	2.18	0.78	4.6	-0.30	-0.07
Domestic cutover	Poor	2.80	1.59	0.57	4.6	4.60	1.00
Forestry	Poor	2.60	1.68	0.65	1.9	NA	NA
Near-natural	Poor	-0.23	-0.33	1.43	92.0	54.70	0.59
Rewetted extraction	Poor	-0.23	-0.23	1.00	92.0	79.80	0.87
Rewetted grassland	Poor	-0.23	0.85	-3.70	92.0	68.10	0.74
Rewetted extraction	Rich	0.50	3.22	6.44	216.0	117.90	0.55

Drained land uses: CO_2 ratio 0.25 to 0.83; the lowest is 4.0x below Tier 1.

Same rows, CH_4 ratio up to 6.30, i.e. 6.3x above Tier 1 for Grassland.

Table S15. Confirmed Annex I saltmarsh in Ireland by the salinity regime of the transitional water body containing it. Salinity is water-column, measured at low-tide seine stations in 2015, not marsh porewater.

Salinity class	Area (ha)	%	Polygons
Not surveyed	1136.8	50.1	1426
Outside any transitional water body	893.3	39.4	3983
Spans 18 ppt	237.8	10.5	658

Confirmed Annex I saltmarsh: 2268 ha. Exposed to the sub-18 ppt default: 238 ha (10.5%).

Unclassifiable for want of salinity data: 1137 ha (50.1%).

Methane penalty where the gate bites: 5.23 t CO_2e ha^{-1} yr^{-1}, against a soil gain of 3.34, so the pathway is a net source of 1.89.

Table S16. Methane from rewetting under three metrics, t CO_2e ha^{-1} yr^{-1}. GWP charges the step up for two decades and then charges only the residual of a now-constant flux.*

Pathway	GWP-100	GWP-20	GWP* years 1-20	GWP* years 21+
Organic soil rewetting	3.312	9.777	8.123	0.927
Irish industrial peat extraction	5.689	16.793	23.175	1.593

A sustained flux under GWP is charged 0.28 times its GWP-100 value once aged past 20 years.*

For the Irish case the methane term falls from 5.69 t CO_2e ha^{-1} yr^{-1} at GWP-100 to 1.59 under GWP beyond year 20, a reduction of 72%.*